Euclidifying Gaia: Classifying Non-Point-Like Sources in the Gaia DR3 Catalogue with a Euclid-Supervised Machine Learning Model

Siwei Zhao | Cambridge Part III Astrophysics Research Project 2025–26
Supervisors: Diana Harrison, Simon Hodgkin, Vasily Belokurov
Institute of Astronomy, University of Cambridge

Abstract

Gaia Data Release 3 (DR3) contains approximately 1.8 billion sources, but its point spread function (PSF) photometry provides limited morphological classification, leaving potentially millions of extended objects unrecognised. The Euclid space telescope offers high-resolution imaging with validated morphological classifiers across two data releases, Early Release Observations (ERO) and Quick Data Release 1 (Q1). This work cross-matches both Euclid releases with Gaia DR3 across their shared sky coverage to train an XGBoost classifier on Gaia parameters. Uniform Manifold Approximation and Projection (UMAP) dimensionality reduction reveals physically interpretable cluster structure in the selected Gaia feature space, confirmed by Euclid imaging cutouts. The classifier achieves galaxy precision of 0.874 ± 0.003 and recall of 0.524 ± 0.005 on the Q1 test set, with the corrected blue and red photometer (BP and RP) flux excess factor contributing 40% of the total discriminatory power. Bayesian recalibration demonstrates that a threshold of $P_{\mathrm{gal}} \geq 0.9$ yields 74–90% galaxy selection purity at the estimated Gaia galaxy fraction of 1–3%. Extending the training set with ERO stars improves purity to 79–92% at the cost of reduced recall, and spatial generalisation tests confirm consistent classifier behaviour across sky positions.

1 Introduction

The European Space Agency's Gaia mission, launched in December 2013, has produced the most comprehensive astrometric survey to date, with Data Release 3 (DR3, June 2022) containing approximately 1.8 billion sources observed across the full sky. Gaia's photometric pipeline is optimised for point sources, employing point spread function (PSF) fitting to extract positions and fluxes. However, the Gaia catalogue inevitably contains potentially millions of extended objects whose morphological properties are poorly characterised by point-source photometry. Identifying these sources requires external labels from surveys with superior imaging, which can then be used to train a classifier deployable across the full Gaia sky.

Given Gaia's brightness limits and angular resolution, the extended sources it detects would not be faint and diffuse but relatively bright and compact objects, including nearby galaxies, low-redshift active galactic nuclei (AGN) and quasi-stellar objects (QSOs) with host galaxy contributions, Galactic nebulae and star-forming regions, and gravitationally lensed systems with detectable arc or multiple-image signatures. Reliably identifying these populations has broad scientific value. QSO classification directly affects the quality of the International Celestial Reference System built on extragalactic sources. Star-forming regions provide insight into ongoing stellar formation and galactic evolution. Gravitationally lensed systems flagged as extended sources provide candidates for cosmological follow-up, as they serve as one of the most powerful probes of dark matter distribution and dark energy in the Universe.

Gaia is uniquely positioned for this classification task. Its catalogue already contains parameters that could encode signatures of source morphology, such as the blue and red photometer (BP/RP) flux excess factor, which is elevated for spatially extended sources, astrometric excess noise, and image parameter determination fit statistics, as described in Section 3.2, meaning that no new observations are required. Furthermore, existing galaxy catalogues from surveys such as the Sloan Digital Sky Survey (SDSS; York et al. 2000), Dark Energy Spectroscopic Instrument (DESI; DESI Collaboration 2024), and Euclid cover only portions of the sky, whereas a Gaia-based extended source catalogue would provide uniform all-sky coverage, including the Galactic plane

and Zone of Avoidance where dedicated galaxy surveys do not observe. Such a catalogue would also serve as a bridge connecting different surveys through Gaia's precise astrometric positions.

Gaia DR3 already includes extragalactic classifications from internal processing pipelines. The Discrete Source Classifier (DSC; Delchambre et al. 2023) assigns classification probabilities for quasars, galaxies, stars, white dwarfs, and physical binary stars using BP/RP spectra, astrometric features, and G-band variability. Bailer-Jones et al. (2023) compile candidate catalogues of 6.6 million quasars and 4.8 million galaxies by combining results from the DSC with additional modules including redshift estimation from BP/RP spectra and surface brightness profile fitting. These classifiers are trained on SDSS spectroscopic labels cross-matched to Gaia sources, identifying sources whose spectra exhibit extragalactic features, achieving purities of 50–70% for complete samples or up to 95% for heavily filtered subsets.

However, these classifications are limited by the properties of their spectroscopic training data. SDSS spectroscopy covers a specific sky footprint and imposes its own magnitude-dependent selection effects, meaning that sources outside these limits are absent from the training set. The Euclid space telescope, with spatial resolution and depth substantially exceeding SDSS's, provides morphological classifications derived from high-resolution imaging independent of spectroscopic target selection, identifying spatially extended sources across different sky regions and to fainter magnitudes. For example, extended Galactic sources such as star-forming regions and nebulae, which spectroscopic classifiers trained on extragalactic labels miss entirely, can be identified morphologically. Similarly, faint or low-surface-brightness sources whose spectra are too noisy for reliable classification may still exhibit detectable PSF-fit residuals revealing spatial extent. Since morphological classification addresses fundamentally different physical properties from spectroscopic classification, Euclid labels provide an independent truth table that complements existing spectroscopic approaches.

This work aims to develop a morphological classifier for Gaia DR3 sources, trained on Euclid imaging labels, that can be deployed across the full sky beyond the Euclid footprint. To this end, we cross-match two Euclid data releases with Gaia DR3: the Early Release Observations (ERO, May 2024) and the Quick Data Release 1 (Q1, March 2025). Using Euclid morphological classifications as training labels, we combine supervised machine learning classification with XGBoost and unsupervised dimensionality reduction with Uniform Manifold Approximation and Projection (UMAP) to distinguish extended from point-like sources in a systematically selected set of Gaia DR3 parameters. Throughout this work, the term "galaxy" is used to denote the positive class in this binary classification, encompassing any source that the Euclid training labels identify as spatially extended, regardless of its astrophysical nature.

We investigate whether Gaia's astrometric and photometric quality parameters encode sufficient morphological information to separate these populations, which parameters carry the strongest signal through systematic feature selection, how the selected features structure in UMAP dimensionality reduction, what classification purity can be achieved at realistic Gaia galaxy fractions, and whether the classifier generalises across sky positions independent of the training footprint.

UMAP analysis reveals physically interpretable cluster structure in the selected feature space, with a distinct galaxy locus at 95% galaxy fraction cleanly separated from the stellar majority, confirmed independently by Euclid imaging cutouts and DESI spectroscopic cross-matching. An interactive 3D UMAP visualisation with per-source Euclid cutouts was developed as part of this work (Appendix B). The XGBoost classifier trained on 238,868 Q1-Gaia sources achieves a galaxy precision of 0.874 ± 0.003 and recall of 0.524 ± 0.005 at the default threshold of $P_{\rm gal} = 0.5$, with the corrected BP/RP excess factor C^* contributing 40% of the total discriminatory power. Bayesian recalibration shows that a threshold of $P_{\rm gal} \geq 0.9$ yields 74–90% purity at realistic Gaia galaxy fractions. Extending the training set with ERO stars at an 8.6:1 star-to-galaxy ratio improves precision to 0.919 ± 0.003 at the cost of reduced recall of 0.454 ± 0.006, increasing recalibrated purity to 79–92% at the same threshold. Spatial generalisation tests confirm consistent predictions across sky positions, and both model configurations are provided for deployment to the full Gaia DR3 catalogue.

The structure of this report is as follows. Section 2 describes the Gaia and Euclid datasets, the cross-matching procedure, and validation of the Euclid morphological classifiers used as training labels. Section 3 details

the methodology, including the XGBoost classification framework, feature selection pipeline, and evaluation framework with Bayesian recalibration. The results are presented in Section 4, covering UMAP cluster analysis with physical interpretation, and XGBoost classifier performance with application to random Gaia sources. Section 5 discusses the implications and limitations of the results. Finally, we summarise the conclusions in Section 6 with directions for future work.

2 Data Processing

This work combines data from two space missions: Gaia, which provides the astrometric and photometric parameters from which classifier features are constructed, and Euclid, which provides morphological classifications used as training labels across two data releases. This section describes each dataset, validates the Euclid classifiers adopted as training labels, details the cross-matching procedure that links them with associated photometric consistency checks, and examines selection effects in the resulting sample.

2.1 Gaia Data Release 3

The Gaia space telescope, launched by the European Space Agency on 19 December 2013 into a Lissajous orbit around the Sun-Earth L2 point, carries a dual-telescope optical instrument with a shared focal plane of 106 CCDs, designed to perform repeated astrometric, photometric, and spectroscopic measurements across the full sky (Gaia Collaboration et al., 2016). DR3, released in June 2022, provides these measurements for approximately 1.8 billion sources (Vallenari et al., 2023). For each source, iterative PSF fitting is performed on the one-dimensional transit profiles collected across multiple scan angles. The astrometric solution delivers positions, parallaxes, and proper motions, while a radial-velocity spectrometer provides line-of-sight velocities for bright sources. Broad-band photometry is produced in three bands: optical G (330–1050 nm), blue photometer $G_{\rm BP}$ (330–680 nm), and red photometer $G_{\rm RP}$ (640–1050 nm), alongside low-resolution BP/RP spectrophotometry (Jordi et al., 2010). Sources that deviate from point-like profiles generate systematic residuals in this fitting process, producing measurable anomalies in the corresponding quality parameters.

All classification features used in this work are drawn from the main `gaia_source` table (Gaia Collaboration, 2023), which provides basic source parameters for every observed source. Depending on data quality and source properties, each source receives a 2-parameter (position only), 5-parameter (position, parallax, and proper motion), or 6-parameter (position, parallax, proper motion and pseudocolour) astrometric solution. Notably, spatially extended sources more frequently receive 2-parameter solutions because their profiles disrupt the astrometric fitting process. The 14 parameters selected for classification comprise dimensionless ratios, significance measures, and colour indices that are largely independent of sky position to first order, making them suitable for training a classifier intended for all-sky deployment. These are listed in Table 2; the systematic selection methodology is described in Section 3.2.

2.2 Euclid

The Euclid space telescope, launched on 1 July 2023 into a large halo orbit around the Sun-Earth L2 point, carries a 1.2 m Korsch-type telescope equipped with two instruments (Mellier et al., 2025). The Visible Camera (VIS) operates in a single broad passband I_E, also referred to as the VIS band, covering 530–920 nm with 0.1 arcsec pixel sampling and a measured PSF full width at half maximum (FWHM) of $0\farcs16$. The Near Infrared Spectrometer and Photometer (NISP) provides imaging in three bands, Y_E, J_E, and H_E, spanning 920–2000 nm at 0.3 arcsec pixel sampling with a PSF FWHM of $0\farcs48$–$0\farcs49$ (Cuillandre et al., 2024). Unlike Gaia, which acquires only one-dimensional strip profiles per transit with limited across-scan resolution, Euclid provides full two-dimensional imaging, enabling direct morphological classification that Gaia's scanning geometry cannot provide. This motivates Euclid's role as the provider of training labels in this work.

2.2.1 Early Release Observations

The ERO data release targeted 17 fields during the performance verification phase from August to December 2023, covering galaxy clusters, nearby galaxies, globular clusters, and star-forming regions with a total of 24 hours of observing time. The resulting science-ready data, stored in the `merge_cat` catalogue, contain approximately 11 million objects in the VIS band and over 5 million common detections in NISP across all fields (Cuillandre et al., 2024).

These catalogues were produced using the SExtractor pipeline (Bertin & Arnouts, 1996), which extracts fluxes using Kron aperture photometry (Kron, 1980), where the aperture adapts its radius to the source light profile using the first moment of the intensity distribution, capturing over 90% of the total flux for both stellar and galaxy profiles. The pipeline provides two morphological classifiers. `class_star` is a neural-network stellarity index ranging from 0 for extended sources to 1 for point-like sources (Bertin & Arnouts, 1996), while `spread_model` is a linear discriminant that quantifies how much a source's profile differs from the local PSF, with values near zero for point sources and positive values for extended sources (Bertin, 2023).

2.2.2 Quick Data Release 1

The Euclid Q1 data release comprises the first routine survey observations from February 2024 onwards, covering three Euclid Deep Fields, EDF-North, EDF-South, and EDF-Fornax, plus the calibration field LDN 1641, totalling approximately 63 deg^2 and 30 million sources (Aussel et al., 2025).

Source detection, multi-wavelength photometry, and morphological measurements are performed by the MERge processing function, which operates on jointly stacked VIS and NISP images and extracts fluxes using both PSF-fitting and Kron aperture methods (Romelli et al., 2025), with results stored in the `mer_final_cat` table of the merged product (MER) catalogue. Morphological classification is provided by the photometric redshift (PHZ) catalogue, which assigns each source a discrete class label `phz_classification` and associated class probabilities such as `phz_star_prob` and `phz_gal_prob`, derived from photometric colour analysis of multi-band photometry (Tucci et al., 2025). The validation of the morphological classifiers from both Euclid data releases is presented in Section 2.4.

2.3 Cross-Survey Magnitude Systems

To enable consistent photometric comparison across surveys, magnitude measurements were selected based on analogous photometric roles. Astronomical magnitudes are defined relative to a reference spectrum, and the surveys considered here use two conventions: the Vega system, which sets its zero-point to the flux of the star Vega, and the AB system, which uses a flat reference spectrum with constant flux density per unit frequency. The two systems differ by a passband-dependent offset.

Gaia DR3: `phot_g_mean_mag` from the `gaia_source` table. PSF-fitted mean magnitude in Gaia's broad G band, serving as the primary photometric reference for all Gaia sources, computed in the Vega system as:

$$\texttt{phot_g_mean_mag} = -2.5 \times \log_{10}(\texttt{phot_g_mean_flux}) + 25.69$$

where 25.69 is the G-band zero-point in the Vega system. For comparison with Euclid AB magnitudes, a correction of $G_{\rm AB} \approx G_{\rm Vega} + 0.11$ mag is applied (Gaia Collaboration, 2023).

Euclid ERO: `v_mag_auto` from the `merge_cat` table. Kron aperture magnitude in the VIS detection image, measured with SExtractor's automatic aperture photometry routine (Bertin & Arnouts, 1996). The magnitude is computed as:

$$\texttt{v_mag_auto} = -2.5 \times \log_{10}(\texttt{v_flux_auto}) + 30.13$$

where 30.13 is the AB zero-point for the instrumental VIS magnitudes (Cuillandre et al., 2024).

EUCLID Q1: `ie_mag` derived from `flux_detection_total` in the `mer_final_cat` table. This is the Kron aperture flux measured in the VIS detection mosaic with `vis_det` = 1, equivalent to SExtractor's `v_flux_auto` (Euclid Consortium, 2025a) and serving as the primary total flux measurement for the PHZ classification pipeline (Tucci et al., 2025). The AB magnitude in the I_E band is computed as:

$$\texttt{ie_mag} = -2.5 \times \log_{10}(\mathrm{ZPC}_{I_E} \times \texttt{flux_detection_total}) + 23.90$$

where 23.90 is the AB zero-point, and $\mathrm{ZPC}_{I_E} = 1.05$ is the VIS band zero-point correction to the AB system.

The two Euclid magnitudes are directly comparable: both represent Kron total flux in their respective detection bands in the AB system. Gaia's G-band magnitude, while produced with PSF fitting rather than aperture photometry, covers 330–1050 nm, fully encompassing the Euclid VIS passband at 530–920 nm as shown in Figure 1, making it a practical reference for photometric consistency checks after applying the AB correction.

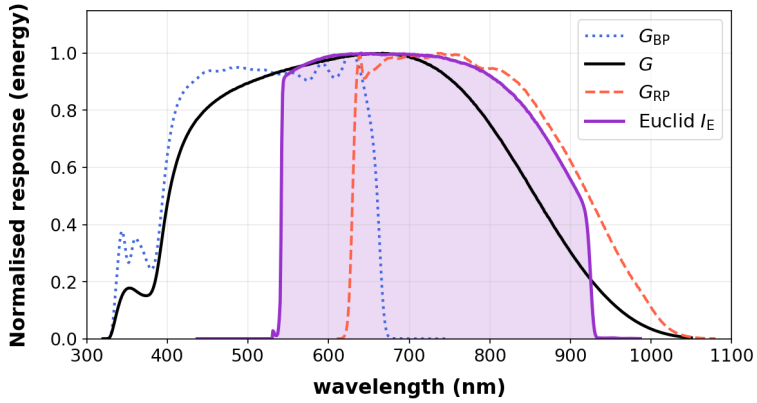


Figure 1: Normalised energy-response curves for Gaia G_{BP} (330–680 nm), G (330–1050 nm), G_{RP} (640–1050 nm) and Euclid I_E (530–920 nm).

2.4 Euclid Classifier Validation

2.4.1 ERO: `class_star` vs `spread_model`

Sanity checks were performed for both ERO morphological classifiers across the full VIS magnitude range.

The `class_star` classifier, with the separation threshold of 0.65 adopted by the ERO pipeline (Cuillandre et al., 2024), shows the expected monotonic behaviour: point-like sources dominate at bright magnitudes with the extended fraction increasing smoothly toward fainter magnitudes, as shown in upper panel of Figure 2.

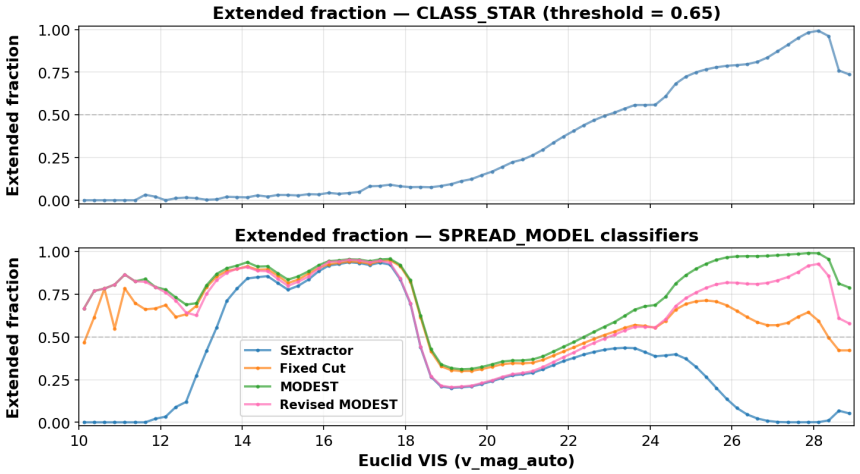


Figure 2: ERO extended source fraction as a function of VIS magnitude for `class_star` and four `spread_model` threshold prescriptions.

The `spread_model` classifier is a linear discriminant comparing each source profile to the local PSF convolved with a circular exponential model with scalelength = 1/16 FWHM, yielding values close to zero for point sources and positive values for extended sources (Mohr et al., 2012). Four threshold prescriptions were tested, listed in Table 1.

Name	Threshold	Reference
SExtractor	$\texttt{SM} > \sqrt{\varepsilon^2 + (\kappa \times \texttt{SME})^2}$, $\varepsilon = 0.005$, $\kappa = 4$	Bertin (2023)
Fixed cut	$\lvert\texttt{SM}\rvert > 0.003$	Desai et al. (2012)
MODEST	$\texttt{SM} + 3 \times \texttt{SME} > 0.003$	Suchyta et al. (2016)
Revised MODEST	$\texttt{SM} + 5/3 \times \texttt{SME} > 0.005$	Drlica-Wagner et al. (2018)

Table 1: `spread_model` threshold prescriptions tested for ERO star-galaxy separation. SM ≡ `spread_model`, SME ≡ `spreaderr_model`.

In contrast to `class_star`, all four threshold variants of `spread_model` exhibit characteristic non-monotonic extended fractions as a function of magnitude, as shown in the lower panel of Figure 2. For the error-weighted prescriptions, since `spreaderr_model` is inversely related to signal-to-noise ratio (SNR), the effective classification boundary shifts with magnitude: at the bright end, high SNR produces a tight threshold that misclassifies genuinely point-like sources with small positive `spread_model` values as extended; at the faint end, low SNR loosens the threshold, allowing extended sources with noise-dominated measurements to fall below the boundary and be misclassified as point-like.

The `class_star` classifier, which does not exhibit these magnitude-dependent artefacts, is therefore preferred over `spread_model` for ERO extended source identification.

2.4.2 Q1: PHZ Classification Validation

The PHZ pipeline assigns classification via a bitmask flag `phz_classification`, where each bit indicates acceptance into a source class: 1 (star), 2 (galaxy), 4 (QSO), and 8 (globular cluster) (Euclid Consortium, 2025b). Single-class sources carry one of these values, while multi-class sources carry bitwise combinations (e.g. 3 = star ∩ galaxy, 5 = star ∩ QSO). Values of -1 and 0 indicate rejected or outlier sources respectively.

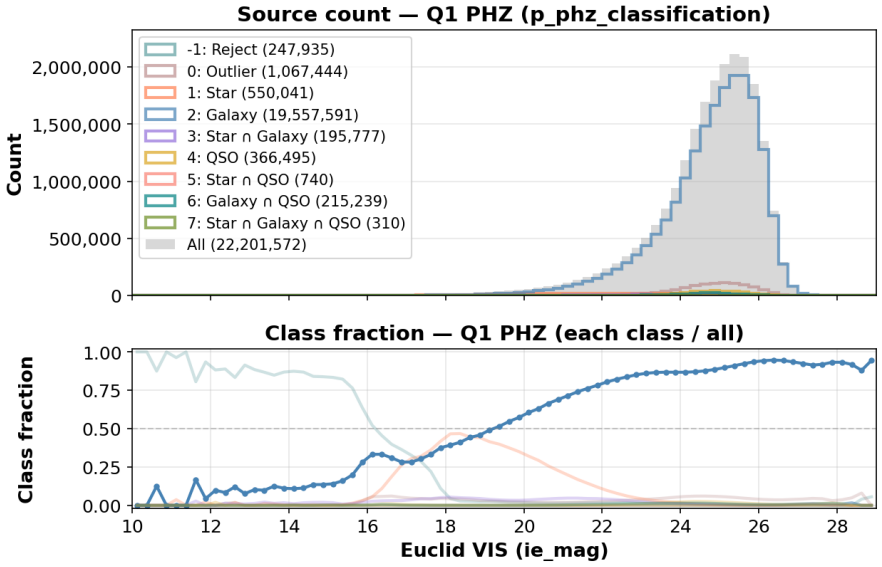


Figure 3: Q1 PHZ classification distributions as a function of VIS magnitude.

The classifications were also validated by examining the distribution of `phz_classification` as a function of `ie_mag`, as shown in Figure 3. Among all sources, the PHZ labels exhibit the expected monotonic behaviour: the galaxy fraction rises steadily toward fainter magnitudes, consistent with the increasing dominance of faint extragalactic sources at the survey depth limit. At mag ≤ 18, rejected sources begin to dominate, reflecting classification uncertainty at the bright end, as the PHZ pipeline is optimised for the faint galaxy population that constitutes Euclid's primary cosmological sample (Tucci et al., 2025).

For use as training labels, only sources with a single-class assignment of `phz_classification` = 1 (star) or `phz_classification` = 2 (galaxy) are retained for the binary classifier, excluding sources classified as QSOs, globular clusters, or multi-class, as well as those rejected by the pipeline.

2.5 Cross-Matching

Cross-matching Gaia DR3 with the two Euclid data releases requires accounting for the proper motion of Gaia sources between the catalogue epoch and the observation epoch, as sources may have moved significantly over the ~7–8 year baseline. For the ~86% of sources with 5- or 6-parameter astrometric solutions, the catalogue proper motions are applied directly. For the remaining 2-parameter sources, which lack proper motion measurements, a proxy is estimated from the median proper motion of at least five nearby 5/6-parameter neighbours within 30 arcsec. In dense stellar fields this recovers the local bulk motion; in sparse fields the uncorrelated neighbour motions cancel to near zero, leaving the position unchanged. All angular separations are computed as exact great-circle distances on the celestial sphere using the haversine formula, consistent with the `q3c_dist` PostgreSQL extension. The canonical cross-match samples adopt a dual-cut criterion: a source is retained only if both the PM-corrected separation and the raw epoch-2016 separation lie within the matching radius. This excludes high proper motion sources whose large positional shift between epochs risks spurious associations or unreliable Euclid photometry due to proper motion blur.

2.5.1 Q1-Gaia Matching

The Q1 pipeline includes pre-computed Gaia DR3 associations stored in the `gaia_id` column, matched within 0.3 arcsec (Romelli et al., 2025). To validate these associations, the per-source proper motion propagation described above was applied independently, with $\Delta t = 8$ yr to a representative Q1 epoch of 2024.0. After correction, 99.9% of the 339,725 matched sources fall within 0.3 arcsec. The corrected angular separation distribution was fitted with a Rayleigh model, restricted to sources below the 90th percentile, as shown in Figure 4. The PM-corrected sources yield $\sigma = 0.015$ arcsec, substantially tighter than the uncorrected distribution at $\sigma = 0.058$ arcsec, confirming both the validity of the pre-computed associations and the 0.3 arcsec matching radius.

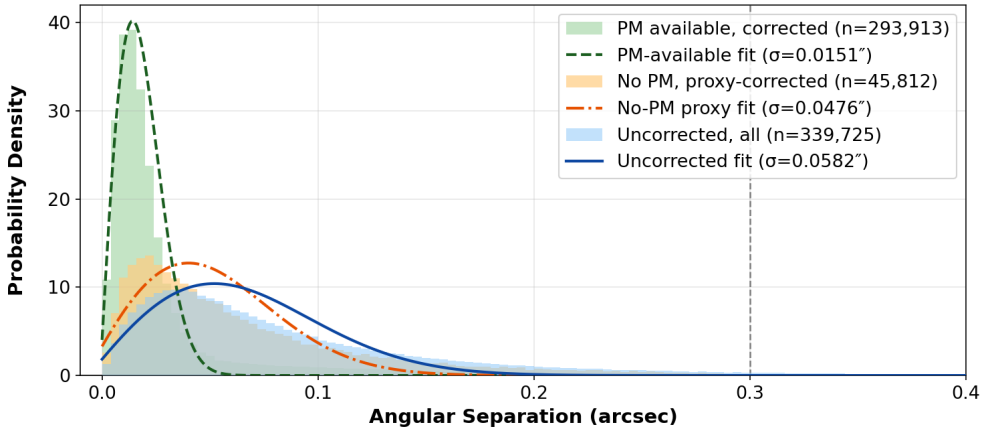


Figure 4: Q1-Gaia angular separation distribution after per-source proper motion correction, with Rayleigh fits for PM-available, proxy-corrected, and uncorrected populations.

2.5.2 ERO-Gaia Matching

As the ERO catalogue does not provide pre-computed Gaia associations, position-based matching was performed independently with one-to-one nearest-neighbour pairing within 0.3 arcsec, using the same per-source proper motion propagation with $\Delta t = 7.5$ yr to a representative ERO epoch of 2023.5. The PM-corrected distribution yields $\sigma = 0.023$ arcsec, compared to $\sigma = 0.037$ arcsec uncorrected, as shown in Figure 5. The neighbour-PM proxy is particularly effective in the dense globular cluster NGC 6397, whose members have a bulk proper motion of $\sim$18 mas yr^{-1} producing an apparent displacement of $\sim$0.12 arcsec over the epoch baseline. This displacement, which appeared as an anomalous secondary bump in the uncorrected separation distribution, is fully absorbed by the proxy correction without requiring field-specific treatment.

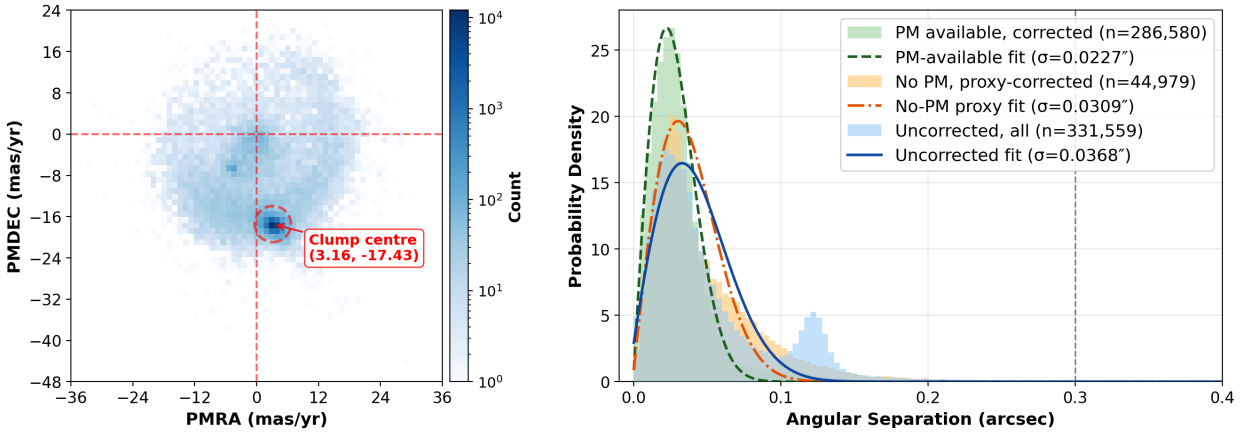


Figure 5: Left: proper motion distribution for ERO-Gaia matches in the NGC 6397 field. Right: ERO-Gaia angular separation distribution after per-source proper motion correction, with Rayleigh fits for PM-available, proxy-corrected, and uncorrected populations.

A pure Rayleigh distribution is not expected in practice, as Gaia's scanning law inevitably produces anisotropic positional uncertainties that vary across the sky. Nonetheless, the model adequately describes the core of both distributions, providing statistical support for the 0.3 arcsec maximum matching distance.

2.5.3 Photometric Consistency Validation

To independently validate the adopted matching radii, magnitude differences $\Delta m = m_{\mathrm{Euclid}} - G_{\mathrm{AB}}$ were examined as a function of angular separation out to 1.0 arcsec. As indicated in Figure 6, the 0.3 arcsec radius captures 86% of the matched sample for ERO-Gaia and 98% for Q1-Gaia. Within 0.3 arcsec, the majority of sources cluster tightly around $\Delta m \approx 0$, confirming photometric consistency for genuine matches.

In the ERO-Gaia sample, sources beyond 0.3 arcsec show increasing scatter and a systematic positive Δm. These are likely cases where extended sources detected by Euclid are spuriously associated with nearby but unrelated Gaia point sources, producing the expected signature of Euclid-fainter magnitudes relative to the brighter mismatched Gaia stars. A maximum matching distance is therefore necessary to exclude such contamination, at the cost of discarding a small fraction of genuine matches that fall beyond the threshold.

The Q1-Gaia distribution reveals a trimodal structure in Δm: a primary concentration at $\Delta m \approx 0$, an upper track at $\Delta m \approx +2$ mag, and a lower track at $\Delta m \approx -3$ mag. The origin of these offsets was diagnosed by inspecting the three Gaia colour indices, as illustrated in Figure 7.

The main track at $\Delta m \approx 0$ is dominated by sources with moderate colour indices, corresponding to typical stellar spectral energy distributions (SEDs) whose flux is concentrated in the 530–920 nm overlap region common to both Gaia G and Euclid $\mathrm{I_E}$ passbands, as shown in Figure 1, producing consistent photometry between the two surveys.

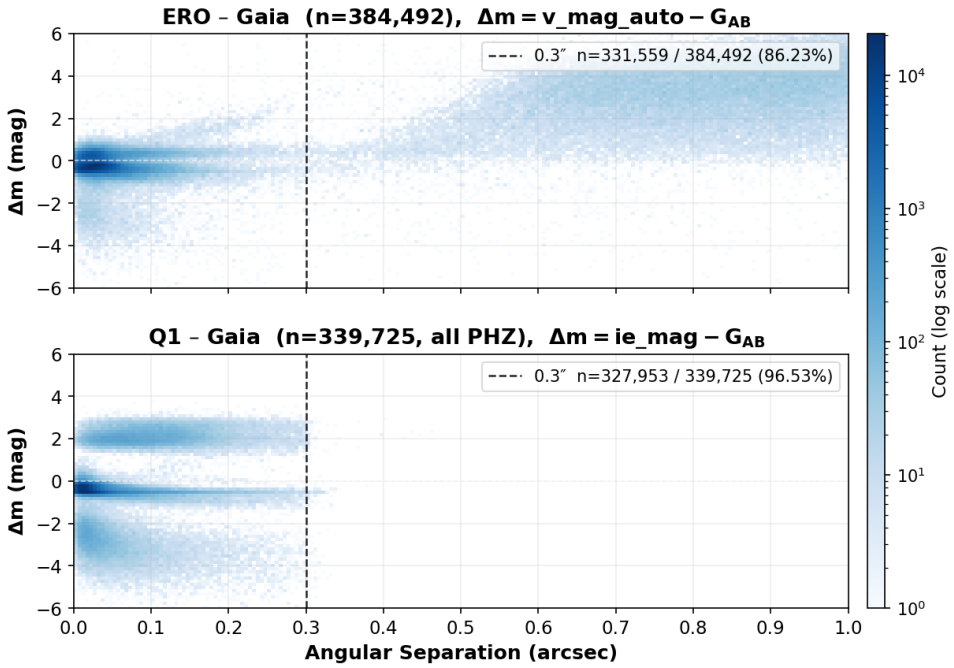


Figure 6: Magnitude difference $m_{\rm Euclid} - G_{\rm AB}$ vs angular separation for ERO-Gaia (top) and Q1-Gaia (bottom).

The upper track at $\Delta m \approx +2$ consists of bright sources. The offset is consistent with pixel saturation in the Euclid VIS detector, which saturates on point sources at $m_{\rm AB} \leq 18.0$ in the nominal 566 s science exposure (Cropper et al., 2025). Saturated pixels in the PSF core lead to an underestimate of the source flux, while Gaia, designed for bright-source photometry, measures these sources accurately, producing the observed Euclid-fainter positive Δm.

The lower track at $\Delta m \approx -3$ exhibits a distinctive colour pattern: $G_{\rm BP} - G < 0$, $G - G_{\rm RP} > 0$, but $G_{\rm BP} - G_{\rm RP} \approx 0$, indicating that both $G_{\rm BP}$ and $G_{\rm RP}$ are systematically brighter than G. This is characteristic of extended sources, and reflects a difference in how Gaia measures photometry across its instruments rather than intrinsic source colour. The G-band magnitude is derived from PSF fitting on the astrometric charge-coupled devices (CCDs), which underestimates the total flux of spatially extended sources, while $G_{\rm BP}$ and $G_{\rm RP}$ are extracted using wider integration windows on the spectrophotometric CCDs that capture more of the extended emission (Riello et al., 2021). Since both $G_{\rm BP}$ and $G_{\rm RP}$ are equally affected, their ratio remains normal with $G_{\rm BP} - G_{\rm RP} \approx 0$. The large negative Δm arises because Euclid's Kron aperture photometry captures the full extended source flux more completely than any of Gaia's extraction methods.

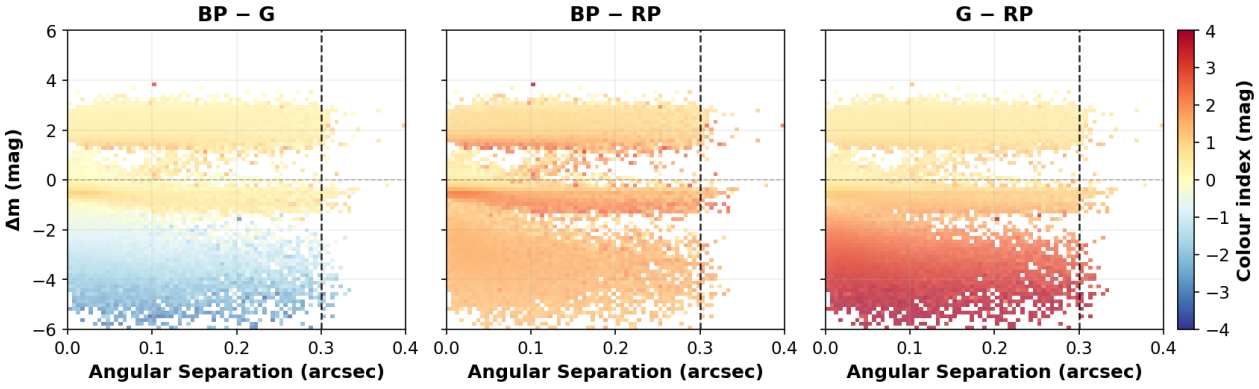


Figure 7: Q1-Gaia magnitude difference vs angular separation, coloured by mean Gaia colour index per bin.

2.6 Training Sample Selection

Figure 8 shows the source counts and class fractions of the Gaia-matched samples as a function of Gaia G. In the Q1-Gaia sample, sources brighter than $G \approx 17$ are predominantly unclassified by the PHZ pipeline, with the unclassified fraction reaching 25–100% below this threshold. The high rejection rate at the bright end may result from saturated photometry compromising the morphological measurements on which the PHZ classifier relies. A magnitude cut of $G \geq 17$ is therefore adopted, keeping the unclassified fraction below 25% and ensuring that training labels are drawn from the regime where the classifiers produce reliable, physically consistent classifications. Since extended sources at these bright magnitudes are exceedingly rare, this cut removes negligible training data for the galaxy class.

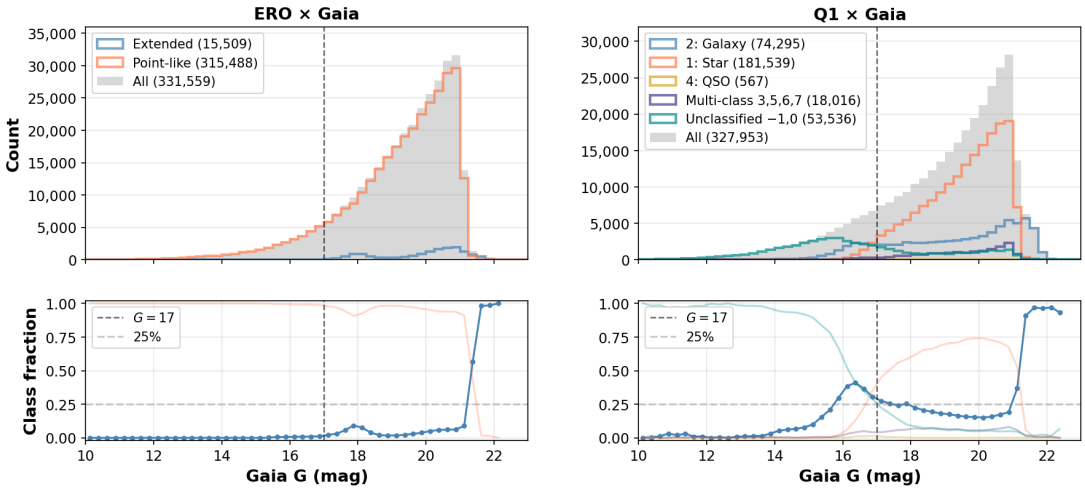


Figure 8: Gaia-matched source counts (top) and class fractions (bottom) as a function of Gaia G magnitude for ERO (left) and Q1 (right).

The ERO-Gaia sample, classified using `class_star`, shows more stable class fractions across all magnitudes with no equivalent bright-end instability, though an anomalous bump at $G \approx 18$ in its magnitude distribution warrants closer examination of the classifier behaviour in this regime.

The 2D distribution of `class_star` vs Gaia G reveals an anomalous concentration of sources at $G \approx 18$, spanning the full range of `class_star` values, as shown in Figure 9, left. This vertical stripe corresponds to the bump seen in the ERO magnitude distribution and suggests a pipeline or calibration artefact affecting a specific magnitude range rather than a physical population. In contrast, the Q1 PHZ galaxy probability shows no equivalent magnitude-dependent concentration, with sources cleanly separated into low and high probability regimes across all magnitudes, as shown in Figure 9, right. The PHZ classifier therefore provides more reliable morphological labels that are less susceptible to magnitude-dependent systematics, and is adopted as the primary source of training labels for the XGBoost classifier.

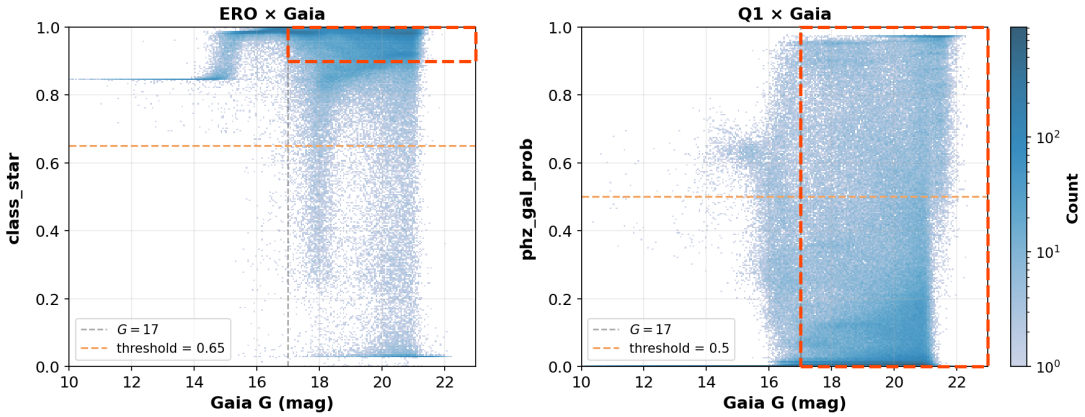


Figure 9: 2D density distributions of morphological classifiers vs Gaia G for the Gaia-matched samples.

However, the resulting Q1-Gaia training sample of 238,868 sources yields a star-to-galaxy ratio of only 2.8:1. Despite the artefacts at intermediate values, ERO sources with `class_star` > 0.9 lie well above the pipeline threshold of 0.65 and are still expected to be predominantly genuine stars, providing an independent pool of 252,171 high-confidence stars from sky regions not covered by the Q1 fields. The Q1-only model is trained first as the baseline, and the effect of extending the training set with these ERO stars is explored in Section 4.2.6.

3 Methodology

3.1 XGBoost Classification

The classifier used in this work is XGBoost (Chen & Guestrin, 2016), a gradient-boosted decision tree algorithm. XGBoost builds an ensemble of shallow decision trees sequentially, with each new tree trained to correct the residual errors of the preceding ensemble. The final prediction is the sum of all trees' outputs, passed through a logistic function to produce a probability as illustrated in Figure 10.

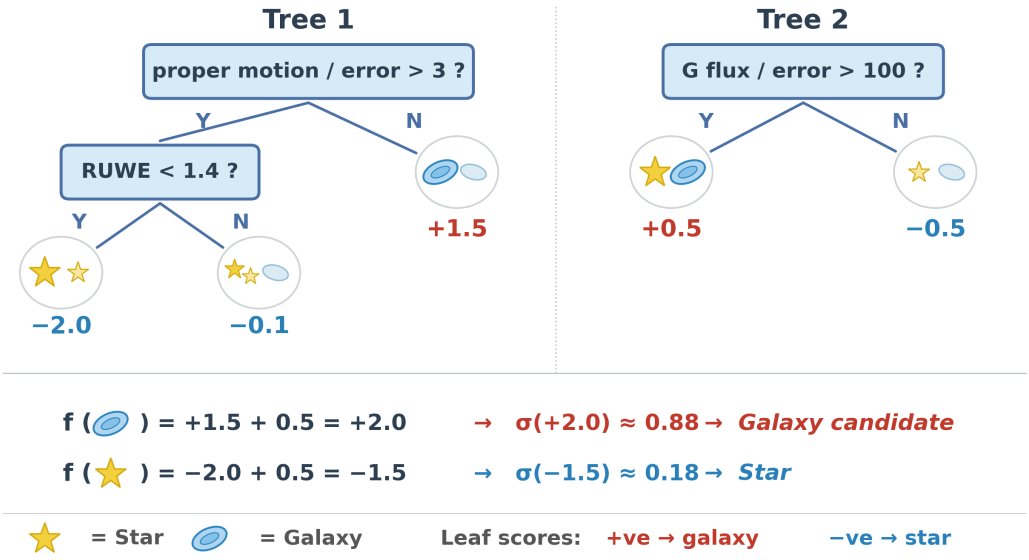
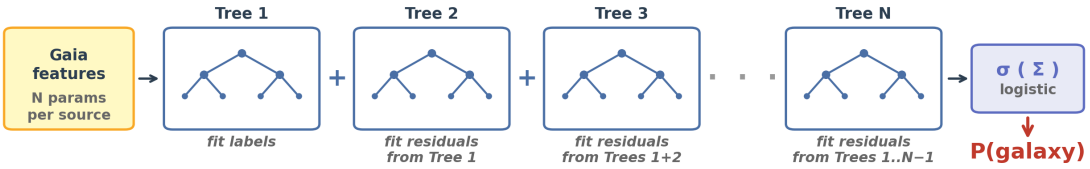


Figure 10: Illustrative example of XGBoost applied to star-galaxy classification.

Each individual tree partitions the input feature space through a series of binary splits, selecting at each node the feature and threshold that best reduces the prediction error, quantified by the gain metric, from among a random subset of the available features. The leaf nodes contain continuous scores that push the prediction toward the galaxy or star class. Because each tree is shallow and captures only part of the signal, the ensemble builds discriminatory power incrementally, with later trees focusing on sources that earlier trees misclassified.



preliminary configuration for feature selection: N = 200 trees, max depth = 8, learning rate η = 0.1, feature subsample = 0.8

Figure 11: Gradient boosting ensemble configuration.

The model was configured as a binary classifier with a logistic objective, a maximum tree depth of 8, up to 200 boosting rounds, a learning rate of 0.1, and 80% random subsampling of both training samples and features

at each round to reduce overfitting. These initial values were later refined through Optuna hyperparameter tuning as described in Section 3.3. The Q1-Gaia cross-matched sample of 238,868 sources with $G \geq 17$ was split into training, validation, and test sets in a 70/15/15 ratio, stratified by class label to preserve the 2.8:1 star-to-galaxy ratio in each subset. The validation set was used for early stopping, halting training if the validation log-loss did not improve for 20 consecutive rounds to prevent overfitting. To assess sensitivity to random initialisation, 10 models were trained with different random seeds, and all reported metrics are ensemble means with uncertainties given by the standard deviation across seeds.

3.2 Feature Selection

The initial candidate feature set comprised 153 columns from the `gaia_source` table spanning source identifiers, sky positions, astrometric quality indicators, photometric measurements and errors, image processing diagnostics, and colour indices. These were systematically reduced through a three-stage selection pipeline.

3.2.1 Coverage Filter

First, parameters with more than 20% missing values across the cross-matched sample were removed, leaving 100 parameters. Retaining only parameters with >80% completeness ensures that the selected features are available for the vast majority of sources in the catalogue.

3.2.2 Physical Inspection and Derived Features

Second, the remaining 100 parameters were inspected for physical relevance and potential circularity. Metadata columns such as source identifiers and reference epochs were discarded. Sky position columns including equatorial, galactic, and ecliptic coordinates were removed, as including these would cause the classifier to learn spatial patterns tied to the training footprint rather than intrinsic source properties. Parameters encoding Gaia's existing DSC classification such as `classprob_dsc_combmod_galaxy` were excluded, as training on these would be circular rather than building an independent classifier from the observational measurements. Absolute flux and magnitude columns were also removed as redundant with the corresponding signal-to-noise ratios and colour indices, which are less sensitive to distance and more informative for morphological classification. Boolean flags, observation counts, and scan direction statistics were similarly excluded as they encode observing strategy rather than source morphology.

In addition, three derived features were computed from existing columns to provide more physically meaningful inputs. Proper motion significance `pm_significance` and positional uncertainty `position_error` were each constructed by combining correlated 2D measurements into single scalar quantities using the full covariance matrix, including the off-diagonal correlation terms `pmra_pmdec_corr` and `ra_dec_corr` respectively, rather than treating the two axes independently. This accounts for the shape and orientation of the uncertainty ellipse in each case, providing more sensitive discriminants than naive 1D signal-to-noise ratios.

The BP/RP excess factor C is defined as the ratio of the sum of the integrated BP and RP fluxes to the broad-band G flux:

$$C = \frac{I_{\mathrm{BP}} + I_{\mathrm{RP}}}{I_G} \tag{1}$$

For a well-behaved point source, this ratio is close to unity after accounting for the different wavelength coverage of the three bands. Extended sources, however, produce systematically elevated values of C because the BP and RP spectrophotometers use wider integration windows than the astrometric CCDs used for G-band PSF photometry, capturing more of the spatially extended emission (Riello et al., 2021). This makes C a direct hardware-level diagnostic of source extent in Gaia.

However, the raw excess factor also varies with source colour even for well-behaved point sources, simply because the three bands cover different wavelength ranges, as illustrated in Figure 1. This makes it difficult to

distinguish colour effects from genuine morphological excess. The corrected BP/RP excess factor C^* removes this colour-dependent baseline following Equation 6 of Riello et al. (2021):

$$C^* = C - f(G_{\rm BP} - G_{\rm RP}) \tag{2}$$

where C is the raw `phot_bp_rp_excess_factor` and f is a piecewise polynomial modelling the expected excess as a function of colour for isolated point sources, with coefficients given in Table 2 of Riello et al. (2021). After this correction, $C^* > 0$ isolates sources with genuine morphological excess independent of their intrinsic colour.

3.2.3 Importance Filter

Third, preliminary XGBoost classifiers were trained using the configuration described in Section 3.1 across multiple trial runs. The gain metric, which measures the cumulative reduction in the loss function from all splits on a given feature across all trees, is used as the feature importance measure to quantify each feature's contribution to the classifier's discriminatory power. Features consistently contributing less than 1% of the total gain were dropped, as were features whose gain showed a standard deviation exceeding 20% of their mean across runs, indicating that their contribution was not stable and may reflect noise rather than genuine discriminatory power.

3.2.4 Final Feature Set

The final feature set comprises 14 parameters (Table 2): six astrometric quality indicators, three photometric signal-to-noise ratios, two colour indices, the corrected BP/RP excess factor, proper motion significance, and positional uncertainty. These are dimensionless ratios, significance measures, and colour indices that are largely independent of sky position to first order, making them suitable for a classifier intended for all-sky deployment.

Parameter	**Physical meaning**
`parallax_over_error`	Parallax significance
`astrometric_gof_al`	Gaussianised astrometric goodness-of-fit
`astrometric_chi2_al`	Unnormalised sum of squared astrometric residuals
`astrometric_excess_noise_sig`	Excess astrometric noise significance
`ipd_gof_harmonic_amplitude`	Image parameter determination goodness-of-fit amplitude
`ruwe`	Renormalised unit weight error
`phot_g_mean_flux_over_error`	G-band photometric SNR
`phot_bp_mean_flux_over_error`	$G_{\rm BP}$-band photometric SNR
`phot_rp_mean_flux_over_error`	$G_{\rm RP}$-band photometric SNR
`bp_g`	$G_{\rm BP} - G$ colour
`g_rp`	$G - G_{\rm RP}$ colour
`corrected_bp_rp_excess`	Colour-corrected BP/RP flux excess C^*
`pm_significance`	2D proper motion significance
`position_error`	Positional error ellipse area

Table 2: The 14 Gaia DR3 parameters used as classifier features.

3.3 Evaluation Framework

Since the classification task treats galaxies as the positive class and stars as the negative class, the two most informative metrics are precision and recall. Precision quantifies the purity of the galaxy selection: of all

sources the classifier labels as galaxies, what fraction are truly galaxies. Recall quantifies completeness: of all true galaxies in the sample, what fraction does the classifier successfully identify. The F1 score is the harmonic mean of the two, penalising models that sacrifice one for the other. The area under the receiver operating characteristic curve (ROC AUC) summarises the classifier's overall discriminatory power across all possible probability thresholds, with 1.0 indicating perfect separation and 0.5 indicating no better than random.

3.3.1 Hyperparameter Optimisation

With the 14-feature set fixed, the initial hyperparameters were refined using Optuna (Akiba et al., 2019), a Bayesian optimisation framework based on the Tree-structured Parzen Estimator sampler. The search explored nine parameters: maximum tree depth, number of estimators, learning rate, subsample ratio, column subsample ratio, minimum child weight, gamma, and L1 and L2 regularisation strengths. Each configuration was evaluated with 5-fold stratified cross-validation on the full training sample. Separate studies of 100 trials each optimised for ROC AUC, galaxy F1, galaxy precision, and galaxy recall.

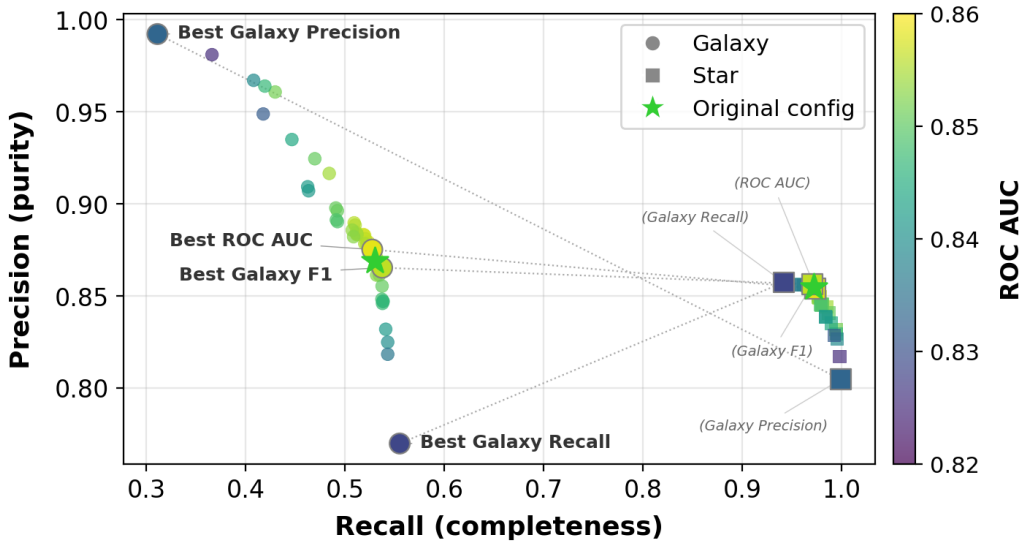


Figure 12: Precision-recall trade-off across hyperparameter configurations. Cloud points show grid search results, bold outlines mark the Optuna-optimised configuration for each of the four target metrics.

The results confirm that classifier performance is largely insensitive to hyperparameter choice, with ROC AUC varying by less than 0.003 among the top-performing configurations, and the full range spanning only 0.04 across all 400 trials including deliberately extreme configurations. As shown in Figure 12, the original configuration sits between the Optuna-optimised ROC AUC and galaxy F1 models, with all three achieving similar performance. The best configurations found by Optuna require substantially more computational cost, typically `n_estimators` $>$ 700 with `learning_rate` $\approx$ 0.02, compared to the original 200 trees at `learning_rate` = 0.1, while producing only marginal improvements of +0.003 in ROC AUC and +0.004 in galaxy F1. The original configuration is therefore retained as it already operates near the performance ceiling for this feature set, at a fraction of the training and inference cost.

3.3.2 Bayesian Recalibration Method

Because the classifier is trained on the Q1 sample where the galaxy fraction is 26%, its output probabilities are calibrated to this training distribution rather than to the true class proportions in the full Gaia catalogue, where galaxies would constitute a much smaller fraction of sources. Bailer-Jones et al. (2023) identify 4.8 million galaxy candidates out of 1.8 billion sources, corresponding to approximately 0.3%, though the true fraction of morphologically extended sources including unidentified galaxies may be higher. A probability

threshold that yields high purity in the training sample will therefore admit more stellar contamination when applied to the full catalogue. To correct for this base-rate mismatch, Bayesian recalibration is applied. For a given probability threshold t, the recalibrated galaxy selection purity is

$$P_{\mathrm{recal}}(\mathrm{gal} \mid P_{\mathrm{gal}} \geq t) = \frac{\mathrm{TPR}(t) \times \pi_{\mathrm{gal}}}{\mathrm{TPR}(t) \times \pi_{\mathrm{gal}} + \mathrm{FPR}(t) \times (1 - \pi_{\mathrm{gal}})} \tag{3}$$

where $\mathrm{TPR}(t)$ is the true positive rate, equivalent to the recall defined above, $\mathrm{FPR}(t)$ is the false positive rate, the fraction of true stars incorrectly classified as galaxies, both measured on the test set at threshold t, and π_{gal} is the assumed true galaxy fraction. This allows the expected purity and contamination of the galaxy selection to be estimated for any assumed galaxy fraction, providing a framework for choosing the appropriate probability threshold when deploying the classifier beyond the Euclid training footprint.

4 Results

4.1 UMAP Space Analysis

UMAP (McInnes et al., 2018) is a nonlinear dimensionality reduction algorithm that projects high-dimensional data into a low-dimensional embedding while preserving both local detail and global structure. Sources with similar behaviours across the feature space are placed close together, while sources that differ are placed further apart. It was applied to the standardised 14 Gaia features of the Q1-Gaia cross-matched sample as a diagnostic tool, with the embedding computed in two dimensions using the default UMAP hyperparameters: Euclidean distance, `n_neighbors` = 15 controlling how many neighbouring points are considered when estimating local manifold structure, and `min_dist` = 0.1 setting how tightly points are packed in the output. Clusters that emerge in this projection represent groups of sources with coherent Gaia signatures, allowing an independent check of whether Gaia features alone can separate stars from extended objects.

Figure 13 shows the UMAP embedding of 50,000 Q1-Gaia sources with $G \geq 17$. Five clusters were identified by Hierarchical Density-Based Spatial Clustering of Applications with Noise (HDBSCAN; McInnes et al. 2017). C0 is a compact, isolated locus at low UMAP 1 with a 95% galaxy fraction. C1 and C2 are isolated islands occupying intermediate positions between C0 and the main stellar body, with galaxy fractions of 16% and 39% respectively. C3 and C4 form the two largest clusters, dominating the right side of the embedding and containing the bulk of the stellar population, with galaxy fractions of only 17% and 27% respectively. Overlaying the PHZ galaxy probability onto the same embedding confirms that the clusters correspond to morphologically distinct populations: C0 is densely populated by high-probability galaxy sources, while C3 and C4 are dominated by sources with low galaxy probability.

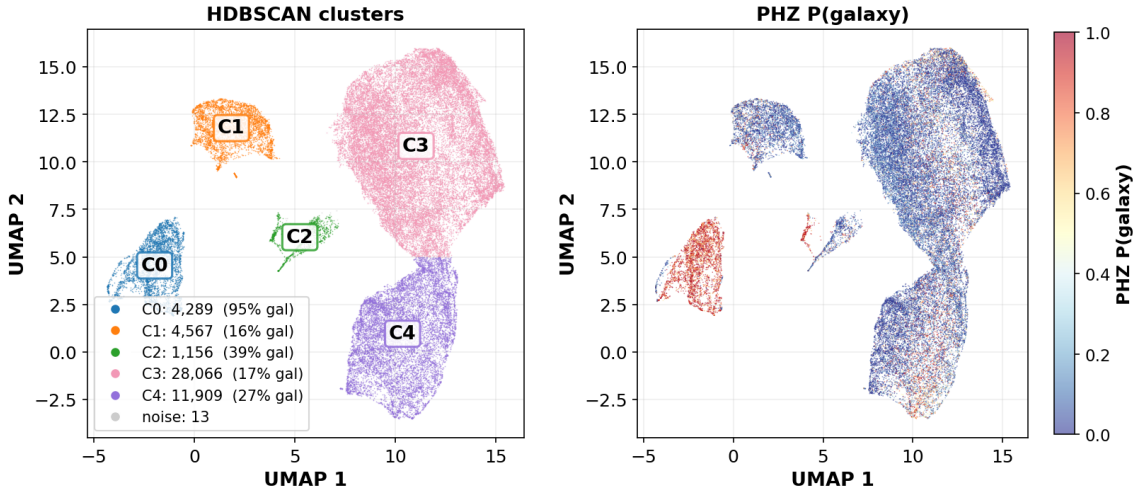


Figure 13: Q1-Gaia UMAP embedding by HDBSCAN cluster (left) and PHZ galaxy probability (right).

To understand why these clusters form, the embedding was coloured by each of the 14 input features, separately for stars and galaxies, with all feature maps provided in Appendix A and explorable in the interactive 3D visualisation at https://euclidifying-gaia.com; see Appendix B. The features naturally separate into groups based on the physical information they encode.

Figure 14 shows the two features that most directly trace spatially extended morphology. The corrected BP/RP excess factor C^* exhibits a strong gradient within C0, ranging from moderate values at the cluster periphery to $C^* \gg 0$ at its core. Positional uncertainty shows a similar gradient within C0, as Gaia's centroiding precision degrades for sources whose light profiles do not match the assumed point-source model. Both features are uniformly low across the stellar clusters C3 and C4, confirming that the morphological distortion is specific to the galaxy locus.

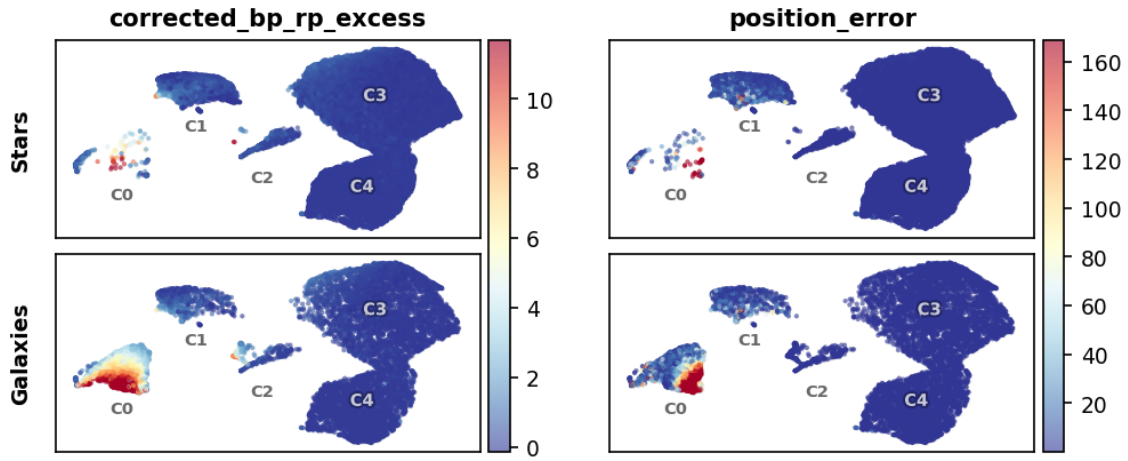


Figure 14: Q1 UMAP embedding coloured by corrected BP/RP excess factor C^* and positional uncertainty.

The three astrometric quality indicators in Table 2 are also elevated in C0, consistent with the point-source astrometric model providing a poor fit to spatially extended galaxies. However, unlike the morphological features above, C2 shows elevated values across all three astrometric indicators as well. Additionally, C2 exhibits the highest renormalised unit weight error (RUWE; Lindegren 2018) values in the embedding. With a galaxy fraction of 39%, C2 likely contains a mixture of compact galaxies marginally resolved by Gaia, QSOs with partially resolved host galaxies, and unresolved stellar binaries. The co-elevation of astrometric residuals in both C0 and C2 suggests a continuum from clearly extended sources to those that are only marginally non-point-like, with C2 representing the transitional regime.

The colour indices $G_{\rm BP} - G$ and $G - G_{\rm RP}$ show smooth gradients across the stellar clusters C3 and C4, tracing the expected variation in stellar spectral type. C0 also exhibits anomalous colour behaviour, as with the morphological and astrometric features above, showing strongly negative $G_{\rm BP} - G$ and strongly positive $G - G_{\rm RP}$, consistent with the wavelength-dependent extraction effects for extended sources identified in Section 2.5.3 rather than intrinsic astrophysical colour.

Figure 15 shows parallax significance, proper motion significance, and G-band photometric SNR across the embedding. Parallax and proper motion measurements are largely absent in C0, indicating these sources received 2-parameter astrometric solutions without parallax or proper motion estimates, consistent with distant extragalactic sources. C1 also lacks parallax and proper motion data for most of its sources, yet has a galaxy fraction of only 16%, suggesting a population of distant stellar sources below the astrometric detection threshold. Within the stellar clusters, the lower periphery of C4 shows elevated values across all three features, identifying a subpopulation of bright, nearby disc stars with well-measured parallaxes, proper motions, and high photometric SNR. The $G_{\rm BP}$ and $G_{\rm RP}$ photometric SNR maps show the same pattern as G-band.

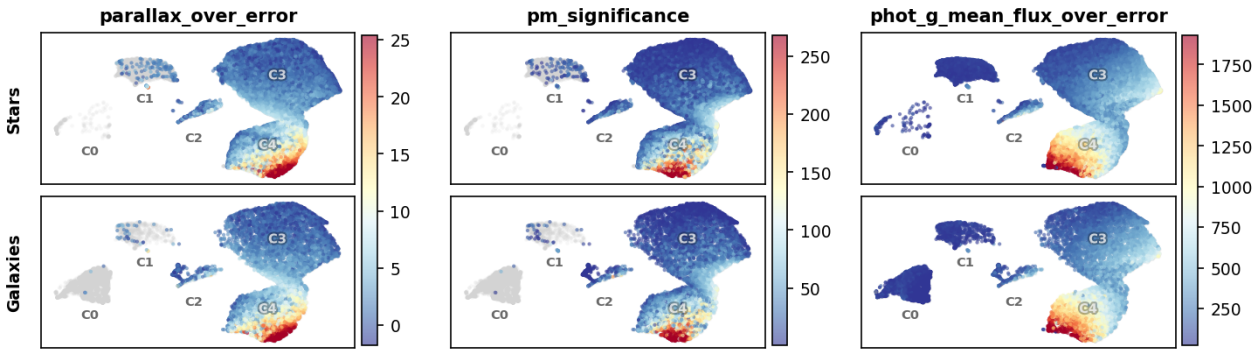


Figure 15: Q1 UMAP embedding coloured by signal-to-noise features. Grey indicates missing data where no measurement was obtained.

Taken together, the feature gradients support the following cluster interpretations. C0 contains clearly resolved galaxies, characterised by high C^*, elevated astrometric residuals, large positional uncertainty, and absent proper motion. C1 contains faint sources below the astrometric detection threshold, predominantly distant stars with a minor galaxy component. C2 represents a transitional population of morphologically ambiguous sources including compact galaxies, QSOs, and astrometric binaries, identifiable by their elevated RUWE and astrometric residuals but lower C^* than C0. C3 and C4 form the main stellar body, separated primarily by photometric SNR and colour, with C4 containing the brightest nearby stars.

These cluster interpretations were confirmed through Euclid cutout images extracted for sources in each cluster, using both VIS-band greyscale imaging and colour composites stacked from VIS, NISP-J, and NISP-H. C0 sources are visually resolved as galaxies with clear extended structure including spiral arms, discs, and diffuse haloes, as shown in Appendix C. C1 sources appear predominantly point-like, consistent with its low galaxy fraction and the interpretation of a distant stellar population below the astrometric detection threshold. C2 sources show a mixture of compact nuclei with faint surrounding emission, marginally resolved galaxies, and close binary pairs, as shown in Appendix C, consistent with the transitional population identified in the feature analysis. Additionally, projecting DESI spectroscopic classifications onto the UMAP embedding confirms that spectroscopically confirmed galaxies concentrate in C0, providing independent validation of the cluster labels. Per-source cutouts can be inspected in the interactive visualisation; see Appendix B.

4.2 XGBoost Classifier Performance

An XGBoost binary classifier was trained on full Q1-Gaia sources with $G \geq 17$ at a 2.8:1 star-to-galaxy ratio, using the 14 Gaia features listed in Table 2 and PHZ star/galaxy labels as training targets. The data were split 70/15/15 into training, validation, and test sets, with the validation set used for early stopping. To assess stability, 10 models were trained with different random seeds; all reported metrics are ensemble means with uncertainties given by the standard deviation across seeds. The inter-seed variation is small, with all metric standard deviations below 0.006, confirming that the classifier performance is robust to random initialisation.

4.2.1 Feature Importance

Figure 16 shows the mean feature importance across all 10 seeds, ranked by XGBoost's gain metric. The corrected BP/RP excess factor C^* dominates with an importance of 0.394 ± 0.016, contributing approximately 40% of the total discriminatory power. This is consistent with the UMAP analysis in Section 4.1, where C^* was identified as the feature most strongly associated with the galaxy locus C0, and with the physical expectation that C^* directly traces source extent as described in Section 3.2.2.

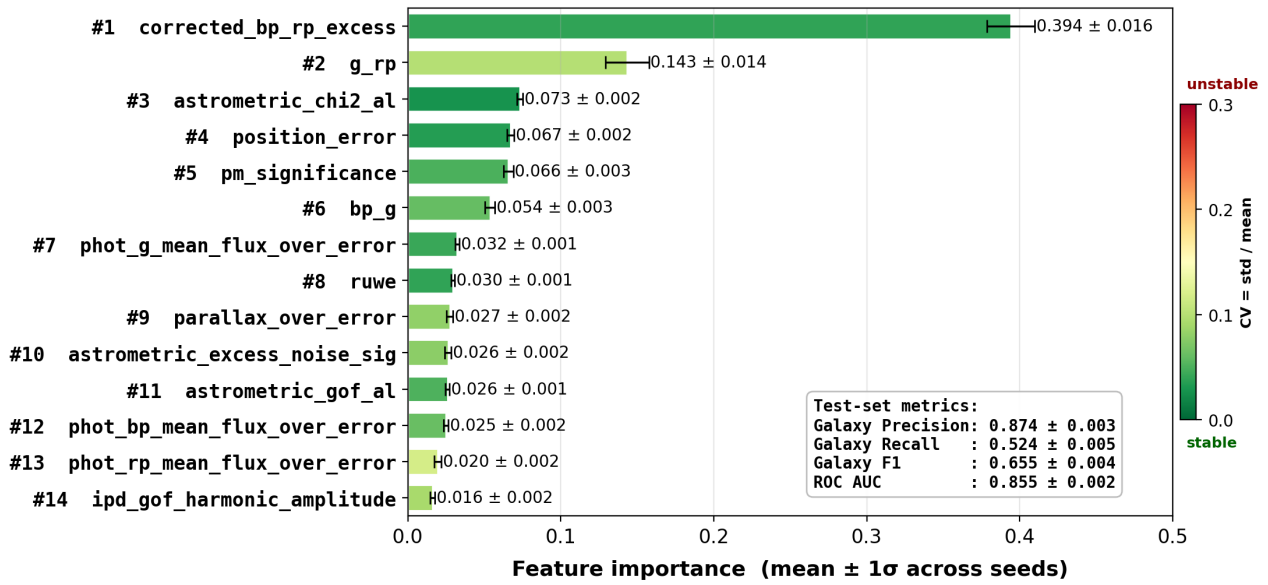


Figure 16: Feature importance for the Q1 model measured by XGBoost gain. Colour indicates the coefficient of variation, with green denoting stable features.

The colour indices $G - G_{\mathrm{RP}}$ and $G_{\mathrm{BP}} - G$ rank second and sixth respectively. As discussed in Section 2.5.3 and Section 3.2, these encode wavelength-dependent extraction effects rather than intrinsic astrophysical colour. Retraining without these two features reduces galaxy recall from 0.524 to 0.489, confirming they provide additional morphological signal. They are retained as the recall improvement outweighs the small precision cost. The 14-feature model is retained because, for the purpose of constructing a galaxy candidate catalogue, recall is the more valuable metric: sources that the model fails to flag as galaxy candidates cannot be recovered by any downstream threshold adjustment, whereas false positives can be removed through the Bayesian recalibration and probability threshold selection described in Section 3.3.2.

The remaining highly ranked features are astrometric and positional quality indicators, reflecting the expected contrast between well-fitted stellar astrometry and the noisy solutions produced by extended or distant extragalactic sources. The model is therefore selecting features that encode physical differences between point-like and extended sources rather than exploiting statistical artefacts. All 14 features have coefficients of variation below 0.13, indicating stable importance rankings across random seeds.

4.2.2 Test-Set Performance

On the held-out Q1-Gaia test set at the default classification threshold of $P_{\mathrm{gal}} = 0.5$, the 10-seed ensemble achieves a galaxy precision of 0.874 ± 0.003, recall of 0.524 ± 0.005, F1 of 0.655 ± 0.004, and ROC AUC of 0.855 ± 0.002, as marked in Figure 16. The inter-seed variation is small, with all metric standard deviations below 0.006, confirming that the classifier performance is robust to random initialisation.

The classifier achieves high purity but limited completeness for the galaxy class: when it labels a source as a galaxy it is correct 87% of the time, but it misses approximately half of the true galaxy population by assigning them sub-threshold probabilities. This is a consequence of the inherent difficulty of identifying extended sources from Gaia parameters alone, as many faint, compact galaxies are photometrically and astrometrically indistinguishable from faint stars.

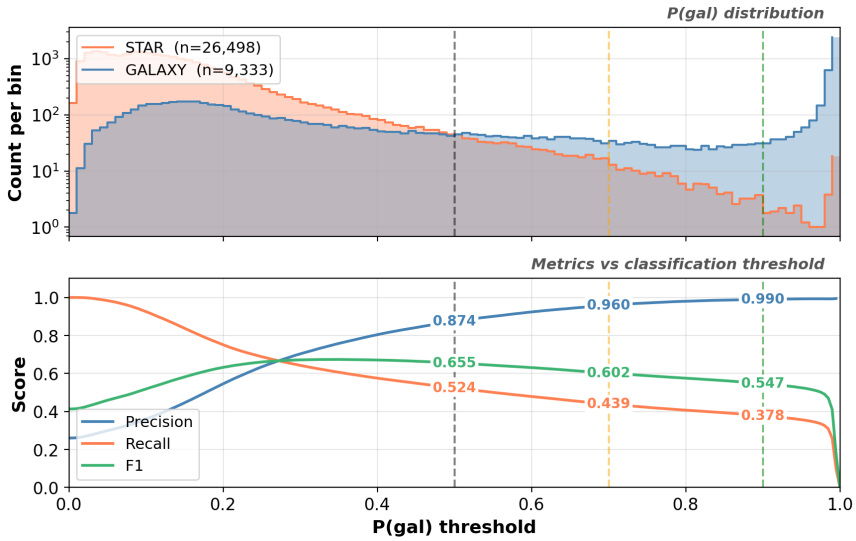


Figure 17: P_{gal} threshold analysis on the Q1 test set. Top: P_{gal} distribution split by true star/galaxy label. Bottom: precision, recall, and F1 as a function of classification threshold.

Figure 17 shows the P_{gal} distributions for stars and galaxies in the test set. The galaxy distribution exhibits a bimodal structure, with a secondary peak near $P_{\text{gal}} \approx 1$ representing the highest-confidence galaxy candidates. This suggests that a higher threshold than the default 0.5 could isolate a high-purity galaxy sample. The lower panel confirms that raising the threshold to 0.9 increases precision from 87% to 99% at the cost of reducing recall from 52% to 38%, a trade-off that may be acceptable for applications requiring a clean galaxy catalogue. The bimodal structure of the galaxy distribution ensures that a substantial population of true galaxies survives even aggressive thresholding. The appropriate threshold choice depends on the deployment context and is quantified through the Bayesian recalibration in Section 4.2.4.

4.2.3 Application to Random Gaia Sources

To assess classifier behaviour on the general Gaia population, the model was applied to 50,000 randomly selected Gaia sources with $G \geq 17$, drawn from across the sky without restriction to the Euclid footprint. Figure 18 shows the resulting P_{gal} distributions. The distribution is broadly declining with the majority of sources at $P_{\text{gal}} < 0.5$, consistent with the expectation that most Gaia sources are stars, and a secondary peak near $P_{\text{gal}} \approx 1.0$ identifying the highest-confidence galaxy candidates. At the most conservative threshold, the predicted galaxy fraction approaches the estimated true Gaia galaxy fraction, suggesting that high-threshold selections may yield catalogues with realistic galaxy counts.

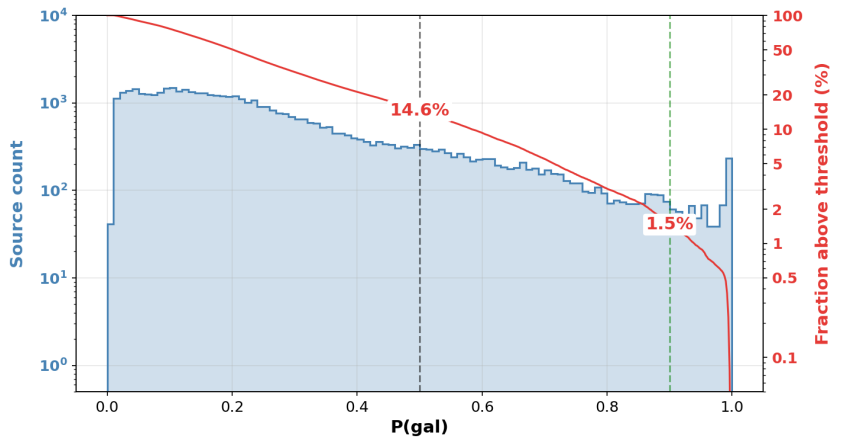


Figure 18: P_{gal} distribution for 50,000 random Gaia sources with $G \geq 17$.

These random Gaia sources were projected into the Q1 UMAP space using the fitted reducer to examine where the classifier's predictions fall relative to the training population structure. Figure 19 shows the result. The density distribution reproduces the overall morphology of the Q1 embedding, with the stellar clusters C3/C4 dominating and the galaxy locus C0 populated at significantly lower density, as expected for Gaia's star-dominated catalogue. Sources coloured by $P_{\rm gal}$ show that high-probability galaxy candidates concentrate in the C0 region, while the stellar body is uniformly assigned low $P_{\rm gal}$, confirming that the classifier is responding to the same feature-space structure revealed by the UMAP analysis.

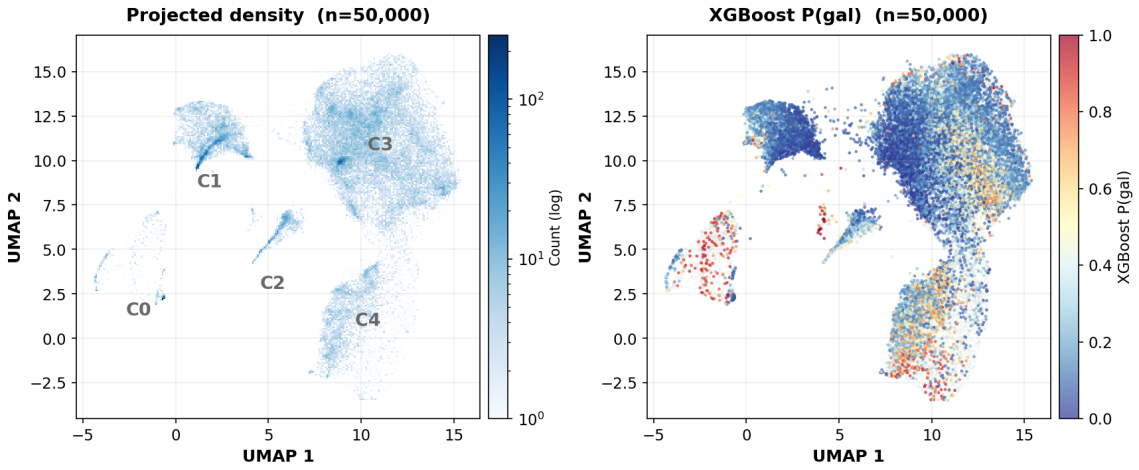


Figure 19: Random Gaia sources projected onto Q1 UMAP space coloured by density (left) and $P_{\rm gal}$ (right).

4.2.4 Bayesian Recalibration

Figure 20 shows the recalibrated galaxy selection purity as a function of the true galaxy fraction for a continuous range of $P_{\rm gal}$ thresholds. At the estimated Gaia galaxy fraction of 1–3%, the default threshold of 0.5 yields a purity of only 17–38%, while raising the threshold to 0.9 improves purity to 74–90%. The right panel presents this as a 2D purity map, allowing appropriate threshold selection for a given scientific application: a threshold at $P_{\rm gal} \geq 0.9$ would yield a fairly pure galaxy sample, while exploratory searches tolerant of contamination may use lower thresholds.

These results demonstrate that while the Q1-only classifier can identify galaxy candidates with reasonable precision in the Euclid footprint, deploying it to the full Gaia catalogue requires higher probability thresholds to maintain acceptable purity. The recalibration framework provides a principled basis for threshold selection depending on whether the application prioritises completeness or purity.

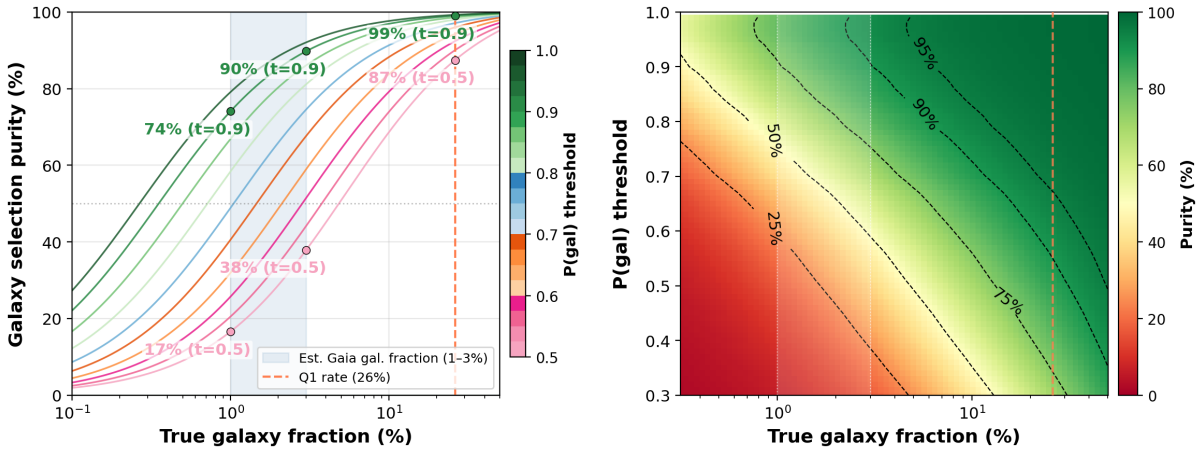


Figure 20: Bayesian recalibration of galaxy selection purity vs true galaxy fraction. Left: purity curves for a range of $P_{\rm gal}$ thresholds. Right: 2D purity map with contour lines.

4.2.5 Spatial Generalisation

A key requirement for all-sky deployment is that the classifier should not encode sky-position-dependent systematics. To assess this, the model was applied to three samples of 50,000 randomly drawn Gaia sources with $G \geq 17$: a spatially unrestricted random sample, a high-latitude subset with $|b| > 10°$ and $|\beta| > 45°$, and a low-latitude subset with $|b| \leq 10°$ or $|\beta| \leq 45°$, where b and β denote Galactic and ecliptic latitude. As shown in Figure 21, the high-latitude cut selects sources within the Gaia scanning-law deep regions where the Q1 training fields are located, while the low-latitude sample draws from the more crowded Galactic plane environment where the classifier will encounter the densest stellar populations.

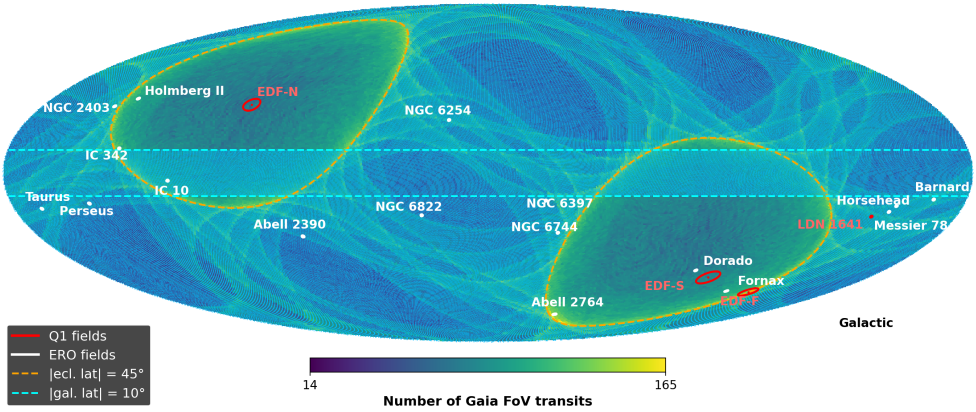


Figure 21: Gaia DR3 scanning law transit density in Galactic coordinates, with Euclid fields marked.

Figure 22 shows the $P_{\rm gal}$ distributions for all three samples. The predicted galaxy fraction at threshold 0.5 is consistent across all three samples at 14.6%, 14.3%, and 14.6% for random, high-latitude, and low-latitude respectively. At the more conservative threshold of 0.9, the high-latitude sample yields a higher galaxy fraction of 3.1% compared to 1.5% for the random sample and 1.3% for the low-latitude sample. This is consistent with a genuine increase in extragalactic source density away from the Galactic plane, combined with improved astrometric precision in the high-transit-density regions, which sharpens the classifier's ability to assign confident high-$P_{\rm gal}$ scores. The low-latitude sample does not produce an excess of false galaxy candidates despite the more crowded stellar environment, confirming that the classifier is not exploiting sky-position-dependent systematics. Projection of all three samples onto the Q1 UMAP embedding confirms that high-$P_{\rm gal}$ sources concentrate in the same galaxy-locus cluster C0 identified in the training data, regardless of sky position.

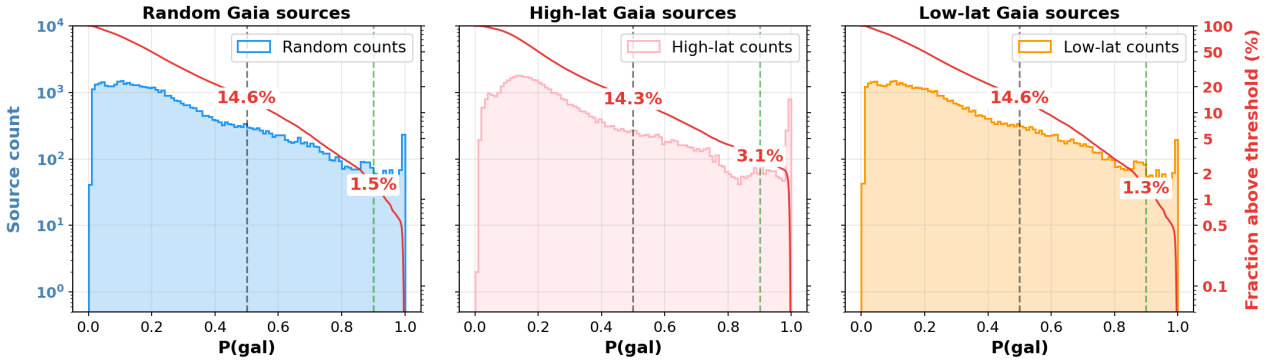


Figure 22: Classifier $P_{\rm gal}$ distributions for random, high-latitude, and low-latitude Gaia samples with $G \geq 17$.

4.2.6 ERO Star Mixing

The Q1 training ratio of 2.8:1 reflects the galaxy-rich Euclid Deep Fields rather than the star-dominated Gaia catalogue where the classifier will be deployed. To test whether exposure to a more diverse stellar population improves star-galaxy discriminations, the training set was extended with high-confidence ERO stars satisfying `class_star` > 0.9 and $G \geq 17$ at controlled ratios. ERO stars were added only to the training split; the validation and test sets remained pure Q1 to ensure metrics are directly comparable.

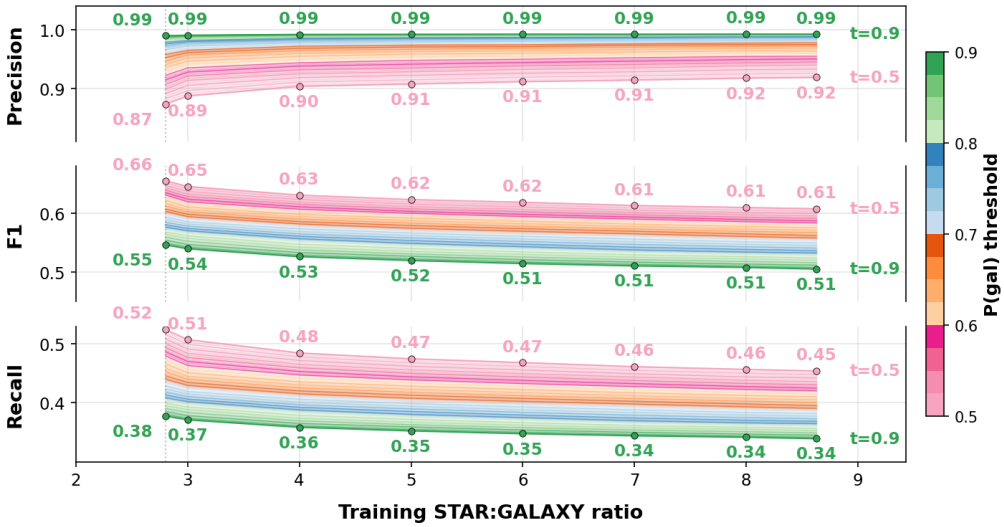


Figure 23: Galaxy precision, F1, and recall vs training star-to-galaxy ratio, coloured by $P_{\rm gal}$ classification threshold. The Q1-only baseline is at ratio 2.8:1.

The star-to-galaxy ratio was swept from the Q1 baseline of 2.8:1 through integer ratios 3:1 to 8:1, up to the maximum achievable ratio of 8.6:1 when all qualifying ERO stars are included. As with the Q1-only model, 10 seeds were trained at each ratio and evaluated on the held-out Q1 test set. Figure 23 shows the resulting precision, recall, and F1 as continuous functions of both training ratio and classification threshold.

Increasing the training ratio monotonically improves galaxy precision at all thresholds, rising from 87% at 2.8:1 to 92% at 8.6:1 for the default threshold of 0.5. This gain comes at the cost of reduced recall, which falls from 52% to 45% over the same range. The F1 score therefore decreases modestly from 0.65 to 0.61. The trade-off is expected: adding more stars to training biases the decision boundary toward higher confidence for galaxy predictions, rejecting borderline sources that would otherwise be classified as galaxies.

The practical consequence of this precision gain is quantified through the Bayesian recalibration framework. Figure 24 shows the recalibrated purity as a function of true galaxy fraction for the Q1-only and 8.6:1 models at three $P_{\rm gal}$ thresholds, with intermediate mixing ratios shown as dotted curves in the left panel. The right panel shows the purity gain of the 8.6:1 model over the Q1-only baseline across the full threshold and galaxy fraction space. The largest absolute gains occur at moderate thresholds and intermediate galaxy fractions, where the baseline model's false positive rate is highest and the additional stellar training data is most beneficial. At low galaxy fractions and high thresholds, the purity gain turns marginally negative, with losses of less than 1 percentage point relative to the Q1-only baseline, as the reduced TPR slightly outweighs the reduced FPR in the recalibration formula. At the estimated Gaia galaxy fraction of 1–3% with threshold 0.9, the 8.6:1 model achieves 79–92% purity compared to 74–90% for the Q1-only model, a gain of up to 5 percentage points.

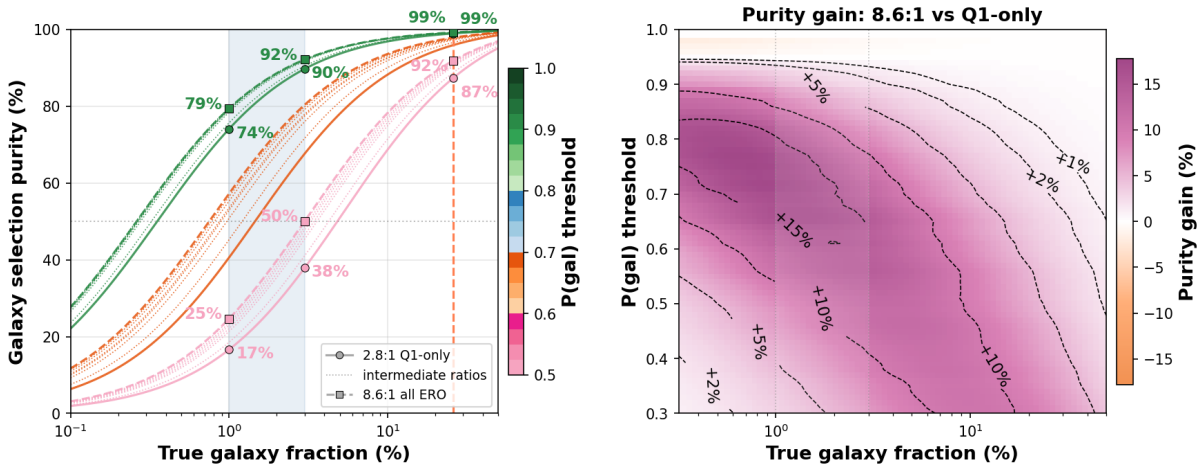


Figure 24: Left: recalibrated galaxy selection purity vs true galaxy fraction for the Q1-only and all ERO mixing models. Right: purity gain of the 8.6:1 model over the Q1-only baseline.

To verify these improvements on the general Gaia population, all models were applied to 50,000 random Gaia sources with $G \geq 17$, as shown in Figure 25. Higher mixing ratios produce progressively sharper bimodal structure, with more sources concentrated at low $P_{\rm gal}$ and fewer at intermediate values, indicating more decisive classifications. At threshold 0.9, the predicted galaxy fraction drops from 1.5% for the Q1-only model to 0.6% for the 8.6:1 model, consistent with either a more selective classifier rejecting marginal candidates or the reduced recall discarding genuine galaxies alongside stellar contaminants.

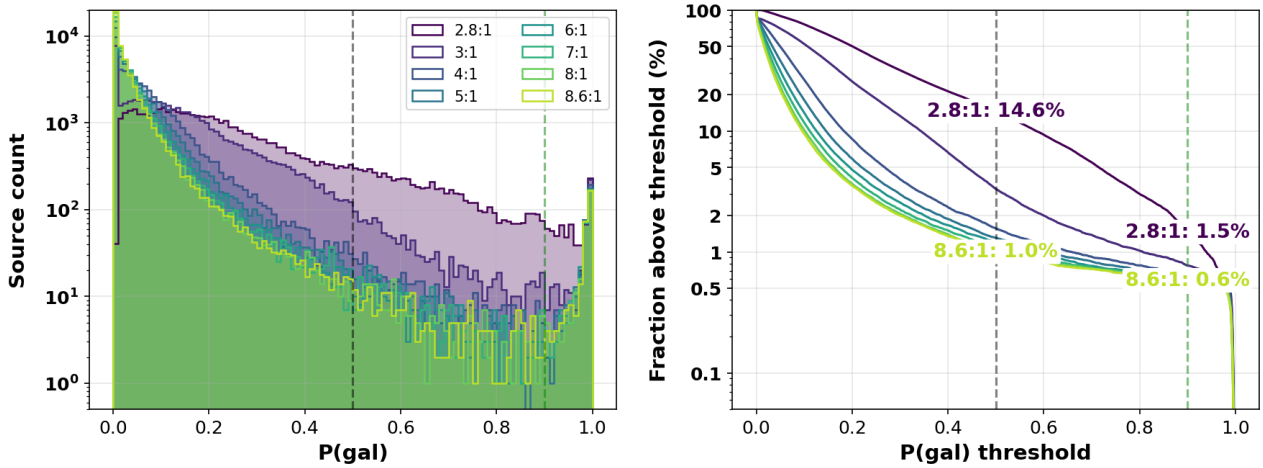


Figure 25: $P_{\rm gal}$ distributions on 50,000 random Gaia sources for all mixing ratios, with cumulative fractions above threshold shown as curves.

ERO star mixing therefore provides a mechanism to improve galaxy selection purity at deployment without retraining on new galaxy labels, by exposing the classifier to a more representative stellar population. The optimal mixing ratio depends on the deployment context: applications requiring high completeness should use the Q1-only model, while applications prioritising catalogue purity benefit from higher mixing ratios combined with conservative $P_{\rm gal}$ thresholds, with the 8.6:1 model at $P_{\rm gal} \geq 0.9$ yielding the highest purity of 92% at an assumed Gaia galaxy fraction of 3%.

5 Discussion

The XGBoost classifier achieves a galaxy precision of 0.874 ± 0.003 and recall of 0.524 ± 0.005 on the Q1 test set, with high purity but limited completeness. The modest improvement from Bayesian hyperparameter optimisation, gaining only +0.003 in galaxy F1 despite exploring 400 configurations across four target metrics, indicates that performance is primarily limited by the discriminative power of the input features rather than the model configuration. A paired t-test across the 10-seed ensemble confirms the improvement is statistically significant at $p = 0.0006$, with the tuned model winning 9 of 10 seeds, but the magnitude gain in galaxy F1 remains small relative to the inter-seed standard deviation of ±0.004, reinforcing that the performance ceiling is largely set by the feature space rather than the choice of model hyperparameters. The Gaia catalogue parameters can reliably identify clearly extended sources whose light profiles produce measurable anomalies in the BP/RP excess factor, astrometric residuals, and positional uncertainty. However, compact galaxies that subtend angular sizes comparable to or smaller than Gaia's PSF produce catalogue parameters that are indistinguishable from those of faint stars. This represents a fundamental ceiling imposed by the resolution of the instrument rather than a limitation of the classification method.

The UMAP analysis reinforces this interpretation. Cluster C0, containing clearly resolved galaxies with $C^* \gg 0$ and elevated astrometric residuals, is cleanly separated from the stellar population and classified with high confidence. In contrast, cluster C2 contains morphologically ambiguous sources including compact galaxies, QSOs with partially resolved host galaxies, and astrometric binaries, all of which produce intermediate $P_{\rm gal}$ values. These sources occupy a transitional regime where the binary star/galaxy classification is inherently uncertain from Gaia parameters alone. A future multi-class extension that treats C2-type sources as a distinct category rather than forcing them into the star or galaxy class could improve both the interpretability and the reliability of the classification.

The spatial generalisation test in Section 4.2.5 provides encouraging evidence that the classifier is not strongly dependent on sky position, with the predicted galaxy fraction at threshold 0.5 consistent across random, high-latitude, and low-latitude samples, and the low-latitude sample producing no excess of false galaxy candidates despite the more crowded stellar environment. However, the training data is drawn entirely from three Euclid Deep Fields, which represent a limited range of Galactic environments. Regions with high stellar crowding, strong extinction, or unusual source populations may present challenges not captured in the training sample: heavy reddening may cause stars to mimic galaxy colours, and dense crowding degrades the astrometric quality features the classifier relies on. The Bayesian recalibration framework mitigates the base-rate mismatch but does not address potential differences in the feature distributions themselves between the training footprint and other sky regions. The per-seed standard deviation provided with each prediction offers a first-order flag for sources in poorly sampled regions of feature space, but validation against a low-latitude spectroscopic reference sample would be needed before treating Galactic plane predictions as reliable.

The ERO star mixing experiment in Section 4.2.6 demonstrates that the precision-recall trade-off can be shifted in favour of purity by exposing the classifier to a more representative stellar population. The 8.6:1 model achieves 79–92% purity at the estimated Gaia galaxy fraction of 1–3% with threshold 0.9, compared to 74–90% for the Q1-only baseline. However, this improvement comes at the cost of reduced recall, with the fraction of true galaxies recovered falling from 52% to 45%. The gains are most pronounced at moderate thresholds and intermediate galaxy fractions, while at high thresholds the purity improvement saturates as both models already reject the majority of stellar contaminants.

The trained ensemble is computationally feasible to deploy across the full Gaia DR3 catalogue. The end-to-end wall-clock is dominated by database I/O rather than inference: query execution and data transfer account for the majority of the runtime, while the 10-seed XGBoost ensemble itself classifies approximately 5×10^4 sources per second on a single CPU core, so the ∼1.5 billion sources with $G \geq 17$ require only ∼8 CPU-hours for inference.

Several additional limitations should be noted. The $G \geq 17$ magnitude cut excludes bright sources where the PHZ classifier produces unreliable labels, but also removes a regime where rare bright galaxies could in

principle be identified. The classifier is trained exclusively on Euclid Q1 PHZ labels, which are themselves derived from photometric analysis rather than spectroscopic confirmation, introducing a dependence on the quality of the training labels. The Bayesian recalibration corrects for the base-rate mismatch between the Q1 training fraction of 26% and the estimated Gaia galaxy fraction of ∼1%, but the true galaxy fraction remains uncertain and varies with sky position and magnitude, meaning that a single recalibration threshold cannot be optimal everywhere.

Additionally, the binary star/galaxy classification adopted here does not distinguish among the different types of extended sources identified as scientifically valuable in Section 1, such as Galactic nebulae, star-forming regions, and gravitationally lensed systems. These would either be absorbed into the galaxy class, if their Gaia parameters resemble those of extragalactic extended sources, or missed entirely if they are excluded by the PHZ training labels which only assign star and galaxy categories. Although the ERO fields include star-forming regions and globular clusters, the `class_star` > 0.9 threshold applied when selecting supplementary ERO stars specifically excludes extended sources, so these Galactic populations do not enter the training set in any form. Recovering these populations would require either a multi-class extension using additional label sources, or post-classification filtering of the galaxy candidate catalogue using spatial cross-matching with known Galactic structures.

6 Conclusions

This work presents a machine learning classifier for identifying extended sources in the Gaia DR3 catalogue using Euclid Q1 morphological classifications as training labels. The main findings are as follows.

The 14 Gaia DR3 parameters selected through coverage filtering, physical inspection, and importance-based pruning provide meaningful separation between point-like and extended sources. UMAP dimensionality reduction reveals physically interpretable cluster structure in this feature space, with a distinct galaxy locus characterised by elevated BP/RP excess, large positional uncertainty, and poor astrometric fits, confirmed independently by Euclid cutout imaging and DESI spectroscopic cross-matching.

The XGBoost classifier trained on 238,868 Q1-Gaia sources achieves a galaxy precision of 0.874 ± 0.003 and recall of 0.524 ± 0.005 at the default threshold of 0.5. The corrected BP/RP excess factor C^* is the dominant feature, contributing 40% of the total discriminatory power, consistent with its role as a direct hardware-level diagnostic of source extent in Gaia. Bayesian recalibration demonstrates that a threshold of $P_{\rm gal} \geq 0.9$ yields 74–90% purity at the estimated Gaia galaxy fraction of 1–3%. Extending the training set with ERO stars at an 8.6:1 ratio improves precision to 0.919 ± 0.003 at the cost of reduced recall of 0.454 ± 0.006, increasing recalibrated purity to 79–92% at the same threshold. The optimal mixing ratio depends on the deployment context, with the Q1-only model preferred for completeness and higher mixing ratios for catalogue purity. Spatial generalisation tests across random, high-latitude, and low-latitude Gaia samples confirm consistent predictions across sky positions, with no excess of false galaxy candidates in the crowded Galactic plane, supporting deployment of both model configurations to the full Gaia sky. The trained Q1-only and ERO-augmented model ensembles, along with usage instructions and the feature specification, are publicly available at https://github.com/SylviaZhao423/euclidifying-gaia-model.

Future Work

Several directions could extend this work. The UMAP cluster structure suggests that a multi-class extension beyond binary star/galaxy classification could be productive. The transitional cluster C2 contains QSOs, compact galaxies, and astrometric binaries that may benefit from dedicated class labels rather than being forced into a binary split. The PHZ pipeline provides additional classification categories beyond star and galaxy, including QSO, multi-class, and rejected labels, which could serve as training targets for such an extension.

Euclid's imaging also reveals additional source types of astrophysical interest within the galaxy class, including star-forming regions and gravitational lensing candidates. Incorporating these as separate labels could enable the classifier to flag scientifically valuable subpopulations within the extended source catalogue.

Finally, the upcoming Euclid DR1 (Euclid Consortium, 2026), covering approximately 1,900 $\deg^2$ of the wide survey, will substantially expand the sky coverage available for training, providing morphological labels across a broader range of Galactic environments. Gaia DR4 (Gaia Collaboration, 2026), based on 5.5 years of observations for approximately 2.8 billion sources, will introduce epoch-level astrometry and photometry for all sources, replacing the summary statistics available in DR3 with per-transit measurements (Brown, 2025). Epoch photometry could reveal variability patterns characteristic of AGN, while epoch astrometry would expose scan-angle-dependent positional shifts diagnostic of spatially extended profiles. The longer time baseline will also improve astrometric precision, sharpening the contrast between extragalactic sources with negligible proper motions and Galactic stars, and improved focal-plane calibrations may reduce systematic effects in the BP/RP excess factor, potentially lowering the detection threshold for the marginally resolved C2-type sources identified in this work. Both releases are expected in late 2026.

Acknowledgements

This work made use of the WSDB database hosted at the Institute of Astronomy, University of Cambridge. All positional cross-matching and indexed range queries were performed using the Q3C PostgreSQL extension (Koposov & Bartunov, 2006).

References

Akiba T., Sano S., Yanase T., Ohta T., Koyama M., 2019, in Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining. pp 2623–2631 (arXiv:1907.10902)

Aussel H., et al., 2025, A&A

Bailer-Jones C. A. L., Teyssier D., Delchambre L., et al., 2023, A&A, 674, A41

Bertin E., 2023, SExtractor Documentation, release 2.24.2, https://sextractor.readthedocs.io/

Bertin E., Arnouts S., 1996, A&AS, 117, 393

Brown A. G. A., 2025. (arXiv:2503.01533)

Chen T., Guestrin C., 2016, in Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining. pp 785–794 (arXiv:1603.02754)

Cropper M. S., et al., 2025, A&A, 697, A2

Cuillandre J.-C., Bertin E., et al., 2024, A&A

DESI Collaboration 2024, AJ, 168, 58

Delchambre L., Bailer-Jones C. A. L., et al., 2023, A&A, 674, A31

Desai S., Armstrong R., Mohr J. J., et al., 2012, ApJ, 757, 83

Drlica-Wagner A., Sevilla-Noarbe I., Rykoff E. S., et al., 2018, ApJS, 235, 33

Euclid Consortium 2025a, MER Photometry Cookbook, Data Product Description v2.0, https://euclid.esac.esa.int/dr/q1/dpdd/merdpd/merphotometrycookbook.html

Euclid Consortium 2025b, PHZ PF Output Catalogue, Data Product Description v2.0, https://euclid.esac.esa.int/dr/q1/dpdd/phzdpd/dpcards/phz_phzpfoutputcatalog.html

Euclid Consortium 2026, Euclid Data Release 1, https://www.cosmos.esa.int/web/euclid/euclid-dr1

Gaia Collaboration 2023, Gaia Data Release 3 Documentation, release 1.3, https://gea.esac.esa.int/archive/documentation/GDR3/

Gaia Collaboration 2026, Gaia Data Release 4, https://www.cosmos.esa.int/web/gaia/data-release-4

Gaia Collaboration Prusti T., de Bruijne J. H. J., et al., 2016, A&A, 595, A1

Jordi C., et al., 2010, A&A, 523, A48

Koposov S., Bartunov O., 2006, in Gabriel C., Arviset C., Ponz D., Solano E., eds, Astronomical Society of the Pacific Conference Series Vol. 351, Astronomical Data Analysis Software and Systems XV. p. 735

Kron R. G., 1980, ApJS, 43, 305

Lindegren L., 2018, Re-normalising the astrometric chi-square in Gaia DR2

McInnes L., Healy J., Astels S., 2017, The Journal of Open Source Software, 2, 205

McInnes L., Healy J., Melville J., 2018, arXiv e-prints

Mellier Y., et al., 2025, A&A, 697, A1

Mohr J. J., Armstrong R., Bertin E., et al., 2012, in Software and Cyberinfrastructure for Astronomy II. p. 84510D (arXiv:1207.3189)

Riello M., et al., 2021, A&A, 649, A3

Romelli E., et al., 2025, A&A

Suchyta E., Huff E. M., Aleksić J., et al., 2016, MNRAS, 457, 786

Tucci M., et al., 2025, A&A

Vallenari A., et al., 2023, A&A, 674, A1

York D. G., et al., 2000, AJ, 120, 1579

A UMAP Feature Maps

Figure 26 shows the Q1 UMAP embedding coloured by all 14 Gaia features used in the classifier, with stars and galaxies separated into upper and lower panels respectively by PHZ classification.

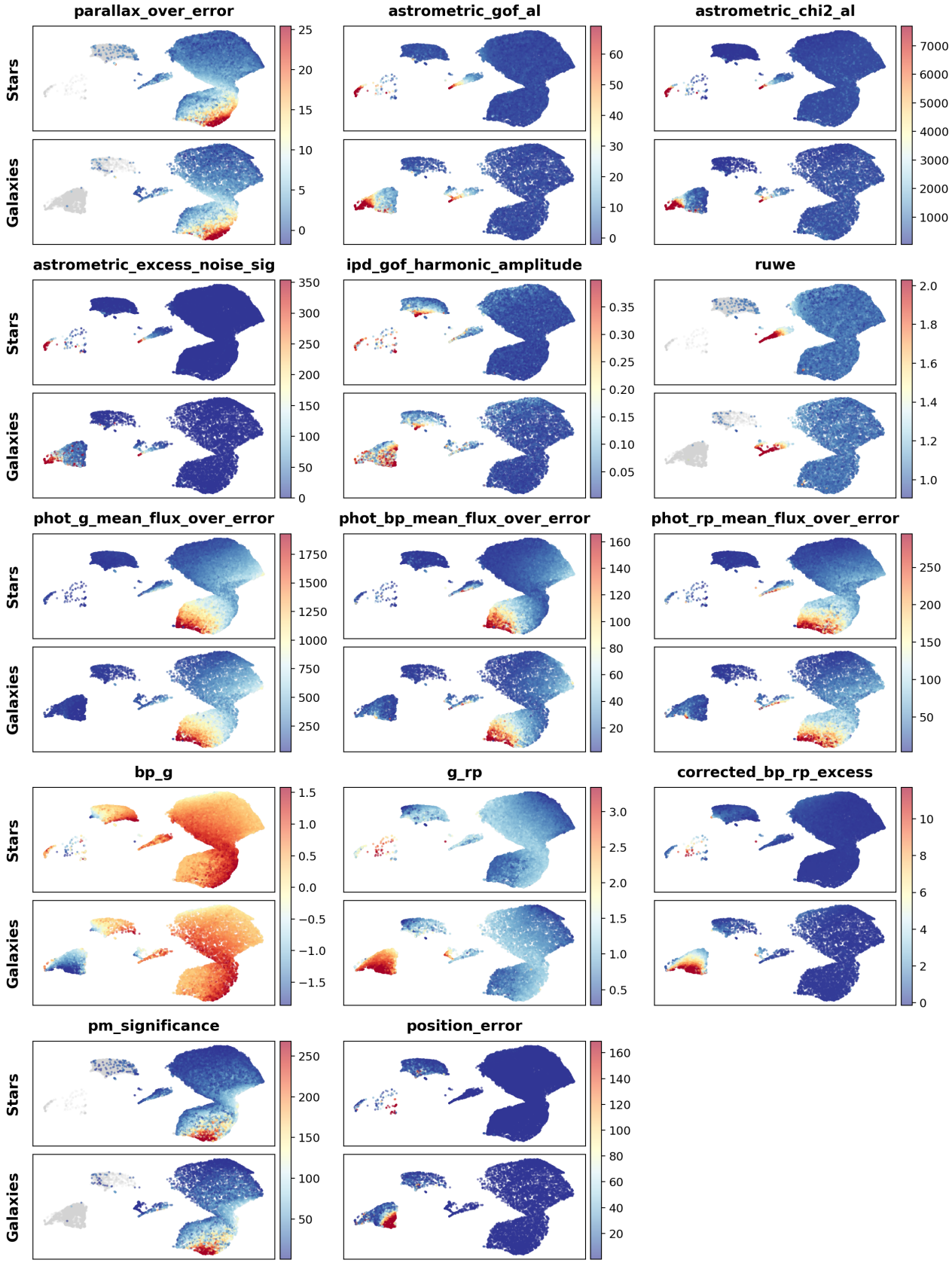


Figure 26: UMAP feature maps for all 14 classifier features.

B Interactive Visualisation

An interactive 3D UMAP visualisation is available at https://euclidifying-gaia.com. The tool displays the UMAP embedding with simultaneous sky position mapping, per-source Euclid cutout inspection, and configurable colour mapping by cluster label, XGBoost P_{gal}, among other source properties. Selecting a cutout sheet in the sidebar opens a dedicated gallery page containing all acquired cutouts for that cluster, with zoom functionality for detailed inspection of individual sources. Figure 27 shows a representative view.

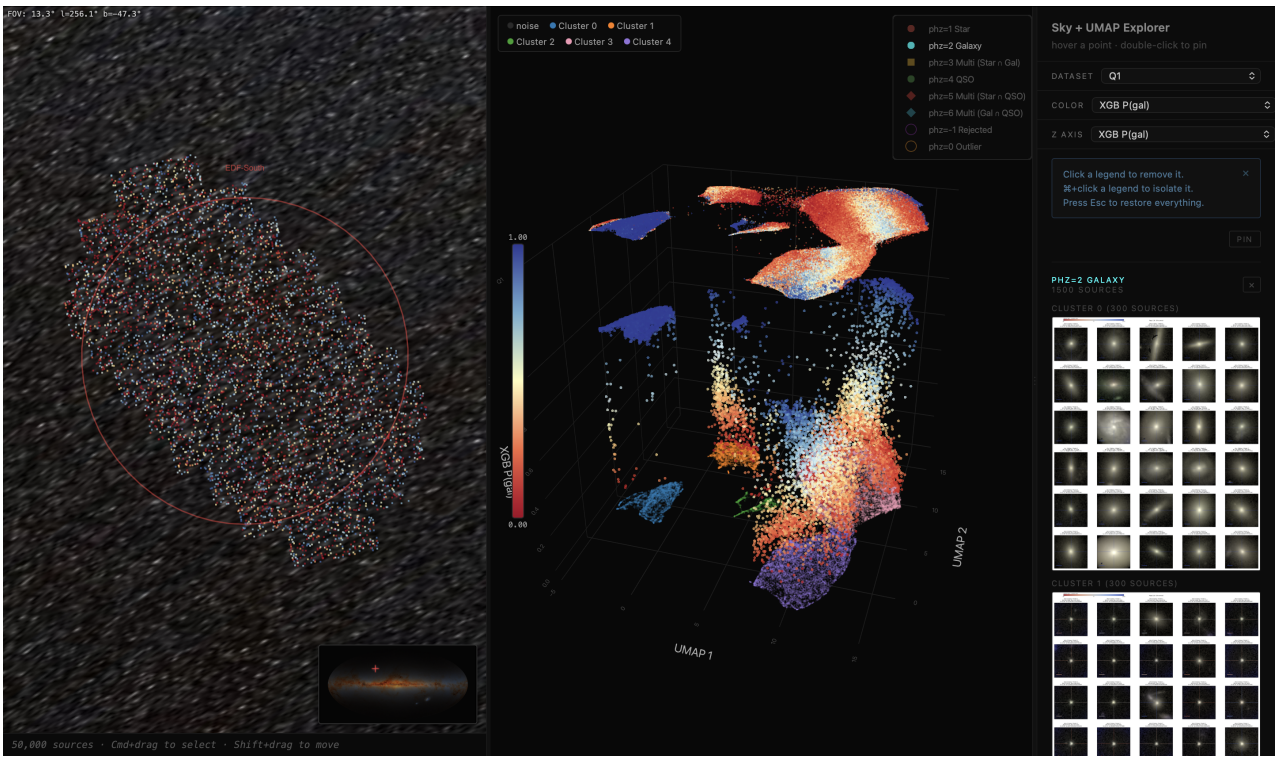


Figure 27: Representative view of the interactive 3D UMAP visualisation.

C Cutout Gallery

Figures 28 and 29 show Euclid colour composite cutouts for sources in UMAP clusters C0 and C2 respectively, cross-matched with DESI spectroscopic confirmations. All sources shown have concordant PHZ and DESI galaxy classifications. Scale bars indicate 1 arcsec.

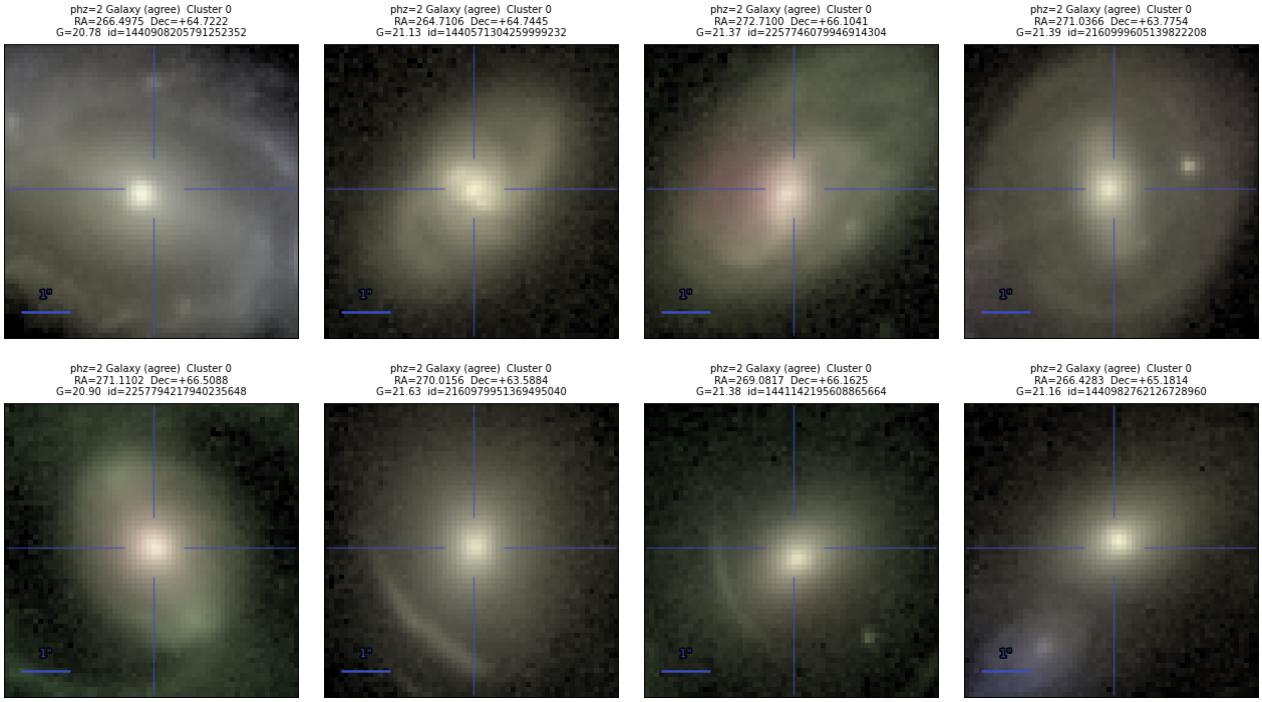


Figure 28: Euclid cutouts for eight C0 sources with concordant PHZ and DESI galaxy classifications.

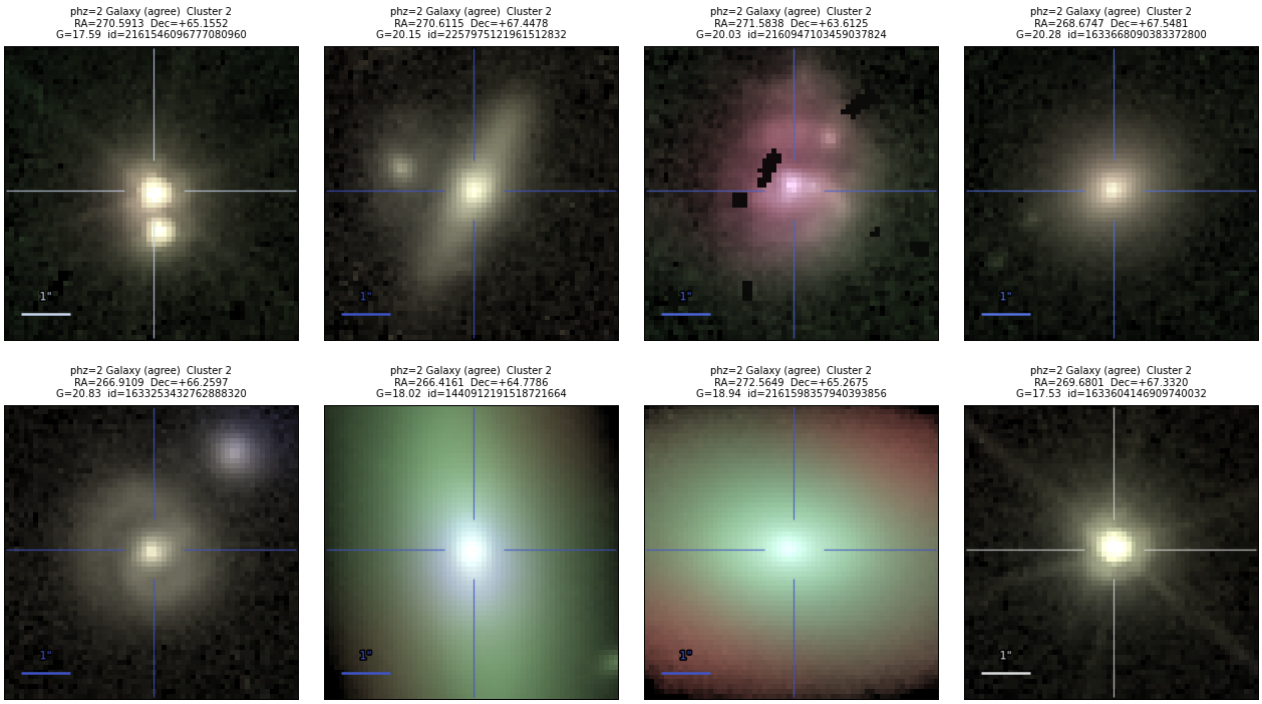


Figure 29: Euclid cutouts for eight C2 sources with concordant PHZ and DESI galaxy classifications.